LEARNER GUIDE · COURSE SLIDES

Agentic AI Automation with n8n

WSQ Course Code: TGS-2023035977

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v48 · 17 Oct 2026

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator will display the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

© 2026 Tertiary Infotech Academy Pte Ltd

YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD — specialises in AI, automation and software engineering.

DELIVERS
WSQ courses on AI agents, automation (n8n) and app development.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

© 2026 Tertiary Infotech Academy Pte Ltd

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with automation or AI tools (if any).
- What you want to automate after this course.

© 2026 Tertiary Infotech Academy Pte Ltd

HOUSEKEEPING

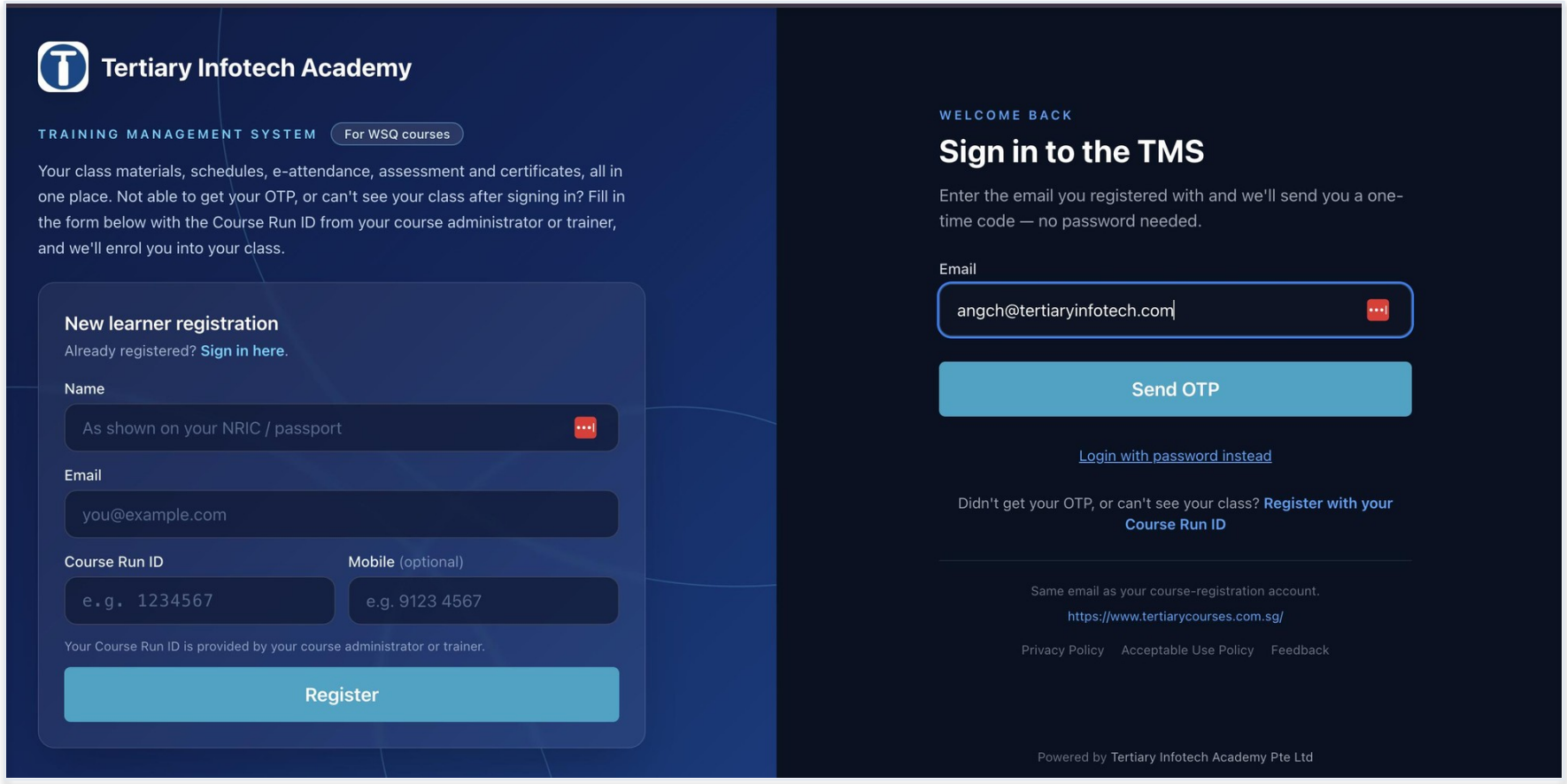
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email + the OTP from your inbox, then download the slides and Learner Guide.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE PORTAL · STEPS

Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course and download the slides + Learner Guide
5. Keep them open — the assessment is open book

© 2026 Tertiary Infotech Academy Pte Ltd

GET THE WORKFLOWS

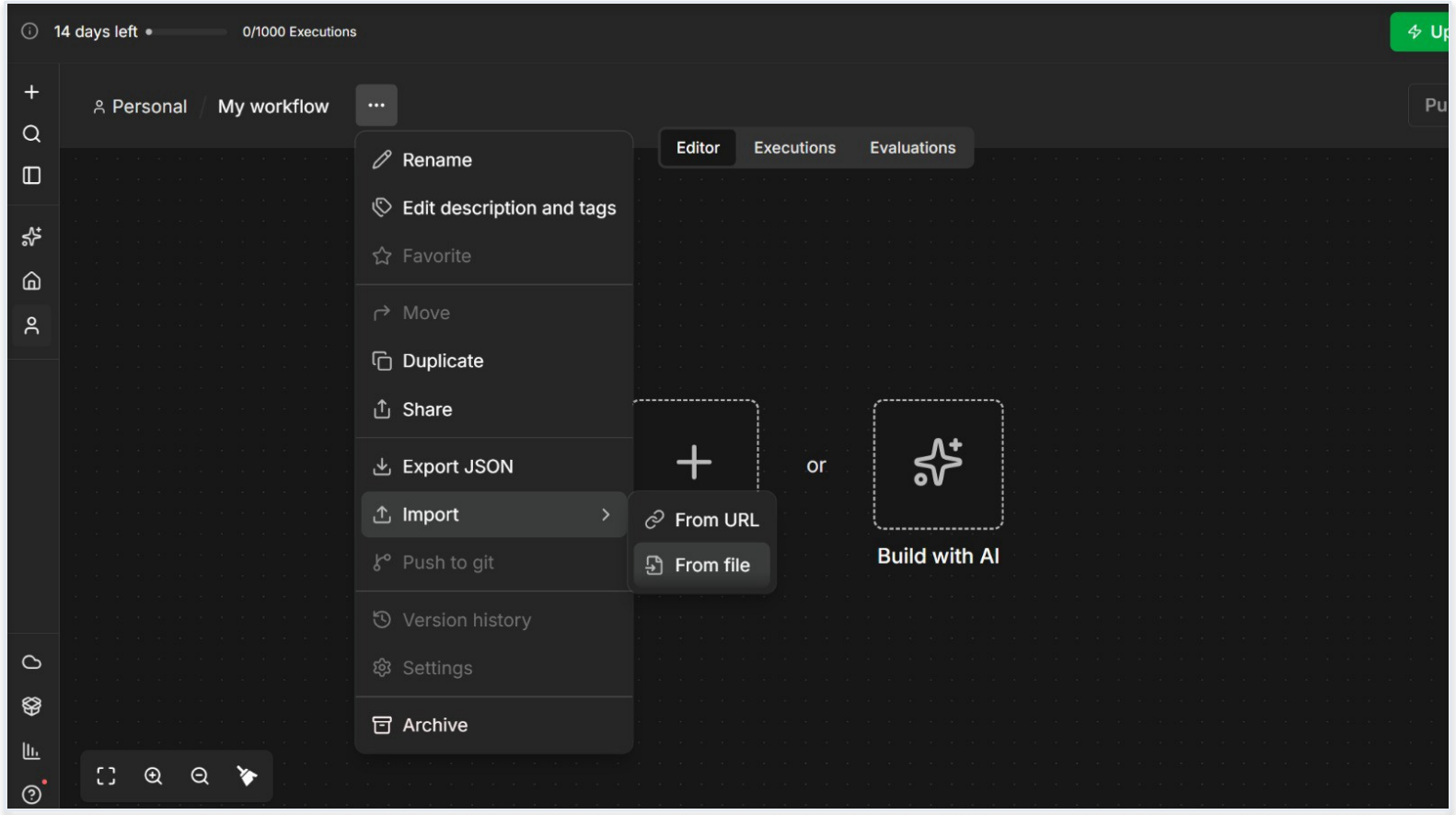
Download the Course Flows (GitHub)

1. Open the course repo: github.com/tertiarycourses/TGS-2023035977-Agentic-AI-Automation-with-n8n
2. Code ▾ → Download ZIP (or git clone) — all workflows are in the labs/ folder.
3. In n8n: Workflows → Add workflow → ... → Import from File → pick the activity .json.
4. After importing, re-select YOUR OWN credentials on each node, then Save & Publish.

© 2026 Tertiary Infotech Academy Pte Ltd

GET THE WORKFLOWS

Import a Workflow from File



Create a NEW workflow, open the '...' menu → Import → From File and pick the activity .json, so your own build is not overwritten.

© 2026 Tertiary Infotech Academy Pte Ltd

SCHEDULE

Lesson Plan — 3 Days, 9:30am–6:30pm

Day 1 — Automate, then add AI

- Topic 1: n8n basics — triggers, actions, data
- Activities 1, 2, 3a, 3b — forms, storage, logic
- Topic 2: AI Agents — model · memory · tools · instructions
- Activities 4a, 4b — Telegram agent + Data Table tool

Day 2 — Connect it to the world

- Topic 3: Webhooks — Activity 5 (website chatbot)
- Topic 4: APIs & HTTP — Activity 6 (day-trading agent)
- Topic 5: RAG — Activities 7a, 7b (grounded agents)

Day 3 — Govern, then prove it

- Topic 6: Security — human-in-the-loop + guardrails
- Activities 8a, 8b, 8c — the HR Service Portal
- Topic 7: Mini capstone + presentations
- Written Assessment (SAQ) 1 hr · Practical (PP) 1 hr

Each day runs 9:30am–6:30pm — 8 instructional hours with a 1-hour lunch; tea breaks counted within.

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL ACHIEVE

Learning Outcomes

1. LO1: Build automated workflows in n8n using triggers, actions, nodes and flows.
2. LO2: Design AI agents with LLMs, memory and tools, triggered from chat / Telegram.
3. LO3: Apply Retrieval-Augmented Generation (RAG) to ground agents in your documents.
4. LO4: Integrate external systems via webhooks, APIs and HTTP requests.
5. LO5: Apply human-in-the-loop and guardrails to secure agentic automations.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE JOURNEY

Twelve Connected Labs

1. Act 1 — Form → Email: your first automation, on a real flyer.
2. Act 2 — Data Table: store every submission, not just forward it.
3. Act 3a/3b — IF logic: route Yes/No to storage or a thank-you.
4. Act 4a — Telegram HR policy agent, built with n8n Assistant.
5. Act 4b — Agent + tool: answers HR questions from a Data Table.
6. Act 5 — Webhook: your agent behind a public website chatbot.
7. Act 6 — APIs: live candles + news → Buy / Sell / Hold calls.
8. Act 7a — RAG: upload a PDF, ask the bot in Telegram.
9. Act 7b — RAG at scale: 20 brochures in a real vector DB.
10. Act 8a–8c — HR portal: approval gate, dashboard, guardrails.
11. Capstone — design and demo your own end-to-end automation.

© 2026 Tertiary Infotech Academy Pte Ltd

BEFORE THE ASSESSMENT

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL ASSESSMENT

Assessment

- Written Assessment (SAQ) — 1 hour.
- Practical Performance (PP) — 1 hour.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A mini capstone project is presented on Day 3.
- An appeal process is available if required.

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

01

TOPIC 1

Workflow Automation with n8n

Why agentic AI · workflow first vs agent first · AI security · what n8n is · sign-in and free AI credits · n8n Assistant · triggers · data nodes

Applied in Activities 1, 2, 3a and 3b

© 2026 Tertiary Infotech Academy Pte Ltd

A Brief History of AI, 2023–2026

Each year moved the skill that matters — from the words you type to the system you build.

Prompt Engineering
2023

▶

Context Engineering
2024

▶

Harness Engineering
2025

▶

AI Agents
2026

Wording the prompt

ChatGPT goes mainstream. Results depend on how well you word the prompt.

Curating the context

Models call tools and MCP launches. Results depend on the documents, data and tools you supply.

Building the harness

Claude Code and Codex: systems that plan, act and check their own work in a loop.

Agents at work

Always-on agents work 24/7 — OpenClaw, Hermes Agent and agent teams take on whole jobs.

From wording one prompt → curating context → engineering the harness → AI agents that work around the clock.

© 2026 Tertiary Infotech Academy Pte Ltd

Generative AI vs Agentic AI

GENERATIVE AI — answers

- You ask; it writes one answer.
- No memory of your systems, no actions.
- Quality depends on what you paste into the prompt.
- Example: ChatGPT summarising a document.

AGENTIC AI — acts

- Works toward a goal, not just one reply.
- Calls tools: looks up data, calls APIs, writes records.
- Holds a conversation with memory.
- Example: a bot that checks the Employees table and answers.

Generative AI produces content. Agentic AI decides and does — which is why it needs governance.

© 2026 Tertiary Infotech Academy Pte Ltd

Agentic AI vs AI Agents

Agentic AI — the broad category

- Any AI system that pursues a goal over many steps: plans, acts through tools, checks the result
- Usually started by a person or a trigger, and stops when the task is done
- Built as a workflow with AI steps, or around an LLM loop
- Includes the orchestration, guardrails and human approvals
- Examples: a Claude Code or Codex session; an n8n workflow with AI steps

AI Agents — always-on agentic AI

- A special type of agentic AI system that exists 24/7
- Usually installed in the cloud, so it serves all the time — no laptop needs to be open
- Waits for events — messages, emails, webhooks, schedules — and acts on them around the clock
- Keeps its memory across conversations and days
- Examples: OpenClaw, Hermes Agent, an n8n agent on a cloud instance behind a webhook

Every AI agent is agentic AI, but not every agentic AI is an AI agent — an AI agent is always on, serving 24/7.

© 2026 Tertiary Infotech Academy Pte Ltd

Workflow First vs Agent First

Workflow first evolves rule-based (RPA) automation: the workflow owns the path, and its steps can be AI or AI agents, as in n8n.

	Workflow First	Agent First
Who decides the path	The workflow — steps may be AI nodes or AI Agent nodes	The agent decides the steps and tools at run time
Behaviour	Deterministic — same input, same path, every time	Probabilistic — the path can change between runs
Efficiency	More effort to design every branch up front	Efficient — one instruction replaces many hand-built branches
Reliability & testing	Easy to test, debug and certify step by step	Harder to test; edge cases surface in production
Cost & speed	An LLM call only where needed — cheap and fast	An LLM call at every step — more tokens, more latency
Best for	Repeatable, regulated, high-volume processes	Open-ended tasks with unpredictable inputs

Source: n8n, In The Loop 2026 — opening keynote and product direction (youtube.com/watch?v=KQRe5UurLQ0)

Workflow first is more deterministic; agent first is more efficient. Companies make as much deterministic as they can.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · THE MODERN VIEW

The Trend: AI-Assisted Workflow Creation

AI-powered coding has flipped the trade-off — the agent's efficiency goes into BUILDING the workflow, not running it.

Describe
the process in plain English

▶

n8n Assistant
or Claude Code / Codex drafts it

▶

You review
test and approve it

▶

Workflow runs
deterministic and logged

▶

AI and agent nodes
where judgement is needed

1 Companies want determinism

Reliability, compliance and audit push businesses to make as much of a process deterministic as possible.

2 AI coding made it cheap

n8n Assistant, Claude Code and Codex are AI agents themselves — they make designing, building and changing workflows fast and cheap.

3 The best of both

An AI agent BUILDS the workflow; a deterministic workflow RUNS the process, with AI and AI Agent nodes where judgement is needed.

The trend: AI-assisted workflow creation — the AI builds the workflow, and the workflow runs the business process.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · COMPLEXITY VS CUSTOMISATION

Where AI Agent Platforms Sit

Three tiers — and AI now helps build at every one of them. This course teaches the middle one.

1 No-code

HOW YOU BUILD

Drag and drop in a fully visual builder.

CUSTOMISATION

Lowest. Fastest route to a working agent, least control over the internals.

2 Low-code · n8n

HOW YOU BUILD

AI-assisted: n8n Assistant, Claude Code or Codex drafts the workflow on a visual canvas.

CODE WHEN NEEDED

Add JavaScript or Python in a Code node where a built-in node is not enough.

CUSTOMISATION

High. 400+ integrations, self-host or cloud. This course's platform.

3 Full-code

HOW YOU BUILD

AI generates the code — Claude Code or Codex write it, instead of coding manually.

CUSTOMISATION

Highest — the most customisable tier. SDKs and frameworks such as LangChain, with real engineering effort to test and run.

Less control, less effort · more customisation, more engineering effort →

AI drafts the workflow in low-code and writes the code in full-code. n8n: analysts build it, engineers extend it.

© 2026 Tertiary Infotech Academy Pte Ltd

Business Value of Workflow-First Agentic Automation

The modern view: what a business gains when AI builds the workflow and the workflow runs the process.

1 Reliable and predictable

The same input follows the same path every time, so the process can be tested and trusted like software.

2 Secure by design

Each step holds only the data and credentials it needs; risky actions wait at an approval gate.

3 Auditable end to end

Every run is logged node by node — you can explain, replay and correct any decision.

4 Lower running cost

Deterministic steps cost nothing in tokens; the LLM is called only where judgement is needed.

5 Faster to build and change

n8n Assistant, Claude Code and Codex draft the workflow in minutes; your team reviews, tests and owns it.

6 Scale around the clock

Triggers respond in real time, 24/7 — ten times the volume costs compute, not ten more hires.

AI where judgement is needed, workflow everywhere else — reliability you can audit, at a cost you can predict.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · USE CASES

Where Agentic Automation Is Used

The same pattern — trigger, agent, tools, oversight — recurs across every function.

1 Customer service

Answer enquiries from your own documents, capture leads, and escalate what needs a person.

2 Finance & accounting

Invoice processing, bank reconciliation, accounts-payable routing and exception handling.

3 Risk & compliance

Screen applications, flag anomalies, and route sensitive cases for human review.

4 HR & people ops

Onboarding, leave applications, policy questions and approval chains.

5 Sales & marketing

Lead qualification, follow-up sequencing, CRM enrichment and content drafting.

6 Operations

Order processing, status notifications, document generation and reporting.

Every one of these is a front door, an agent, some tools and a decision about where the human belongs.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · AI SECURITY

5 Principles of AI Security

Agents act on your behalf. These five controls keep what they can see and do safe, reviewable and accountable.

1

Data Control

WHAT AI CAN SEE

Classify your data — public, internal, confidential — and decide exactly which data the AI may access.

2

Permission Control

WHAT AI CAN DO

Grant only the rights needed — read, write, edit, delete. Ask before anything destructive.

3

Human in the Loop

WHO APPROVES

A person reviews and approves high-impact actions — payments, deletions, customer emails.

4

Traceability

WHAT HAPPENED

Log every prompt, tool call and change so actions can be audited, explained and rolled back.

5

Right-Sized Models

NOT THE MOST POWERFUL

The smallest model that does the job carries less autonomy and less risk than the most powerful one.

In n8n: scoped credentials per node, approval gates (Send and Wait), the Executions log, and a small model per AI step.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · AI SECURITY

Workflow First vs Agent First — AI Security

A fixed path limits the blast radius: a manipulated AI step cannot reach tools that are not wired to it.

Principle	Workflow First	Agent First
Data control	Each node sees only the fields wired into it	The agent can reach any data its tools expose
Permission control	Each credential is scoped to one node and one action	The agent holds every tool's rights at once
Human in the loop	The approval gate sits at a fixed, designed step	Depends on the agent choosing to ask
Traceability	Every run is logged node by node in Executions	Reasoning varies run to run and is harder to explain
Right-sized models	A small model for each narrow AI step	One capable model with broad autonomy

The path is fixed, so every control is designed in rather than hoped for — that is why regulated processes stay workflow first.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · WORKFLOW AUTOMATION

What Is n8n?

A workflow-first agentic AI platform: the workflow owns the path, and AI and AI Agent nodes sit inside it.

1 Build visually

Drag nodes onto a canvas and wire them together. 400+ app integrations, plus HTTP and Code nodes when no integration exists.

2 Run anywhere

Free cloud trial at n8n.io, or self-host with Docker. Webhooks, schedules and chat triggers work the same in both.

3 AI-native

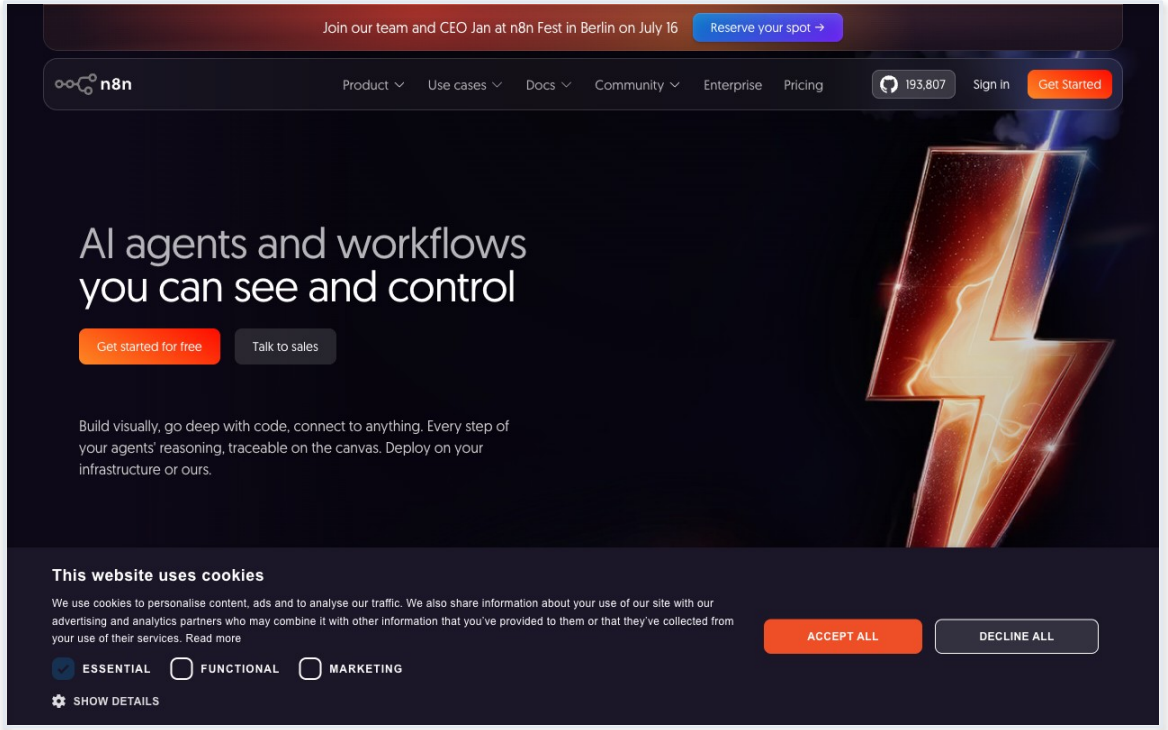
AI Agent and LangChain nodes are built in — attach any LLM (OpenAI, Gemini, Claude), memory, tools and vector stores.

n8n is workflow-first agentic AI — deterministic workflows with AI and agents built in, on one canvas.

© 2026 Tertiary Infotech Academy Pte Ltd

n8n.io

- n8n is an open, fair-code workflow automation platform.
- Free cloud trial at n8n.io, or self-host with Docker.
- Same visual editor in both.
- This course: start on the cloud trial, keep the Docker option for after.



© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · WORKFLOW AUTOMATION

What Automation Removes

Automation is not 'the same work faster' — it removes the hand-offs where a person re-types what another system already knows.

MANUAL

Form filled in ▶ Someone reads it ▶ Re-typed into Excel ▶ Someone remembers to reply

AUTOMATED

Form submitted ▶ Flow triggered ▶ Row written ▶ Reply sent

Consistency

The same input produces the same output, every time, at 3am.

Traceability

Every run leaves an execution history someone can read a year later.

Capacity

People move up to judgement calls; the machine does the re-typing.

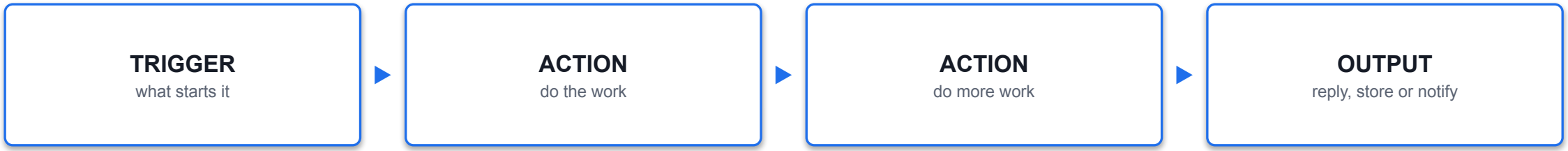
Automate the hand-offs first — that is where time and errors live.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · WORKFLOW AUTOMATION

Every Workflow Is the Same Shape

Trigger first, then actions in order, with data flowing left to right. Only the nodes change.



1 One trigger

A workflow has exactly one entry point. Change the trigger and you have a different flow — the actions may not change at all.

2 Actions are ordered

Each action runs after the one before it and can read the outputs of every node before it.

3 Data flows as JSON

Every node passes a list of JSON items to the next — the same structure in every lab.

Trigger, actions, output. Learn this once in Activity 1 and every later lab is a variation.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · TRAINING ACCOUNTS

n8n Training Accounts

Each learner signs in to n8n with ONE assigned account. Use only the account your trainer gives you.

#	Account (email)	#	Account (email)
1	n8n18oct1@tertiaryinfotech.com	7	n8n18oct7@tertiaryinfotech.com
2	n8n18oct2@tertiaryinfotech.com	8	n8n18oct8@tertiaryinfotech.com
3	n8n18oct3@tertiaryinfotech.com	9	n8n18oct9@tertiaryinfotech.com
4	n8n18oct4@tertiaryinfotech.com	10	n8n18oct10@tertiaryinfotech.com
5	n8n18oct5@tertiaryinfotech.com	11	n8n18oct11@tertiaryinfotech.com
6	n8n18oct6@tertiaryinfotech.com	12	n8n18oct12@tertiaryinfotech.com

Password: provided by your trainer in class · Do not change the password or save real credentials in it.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · SIGN IN

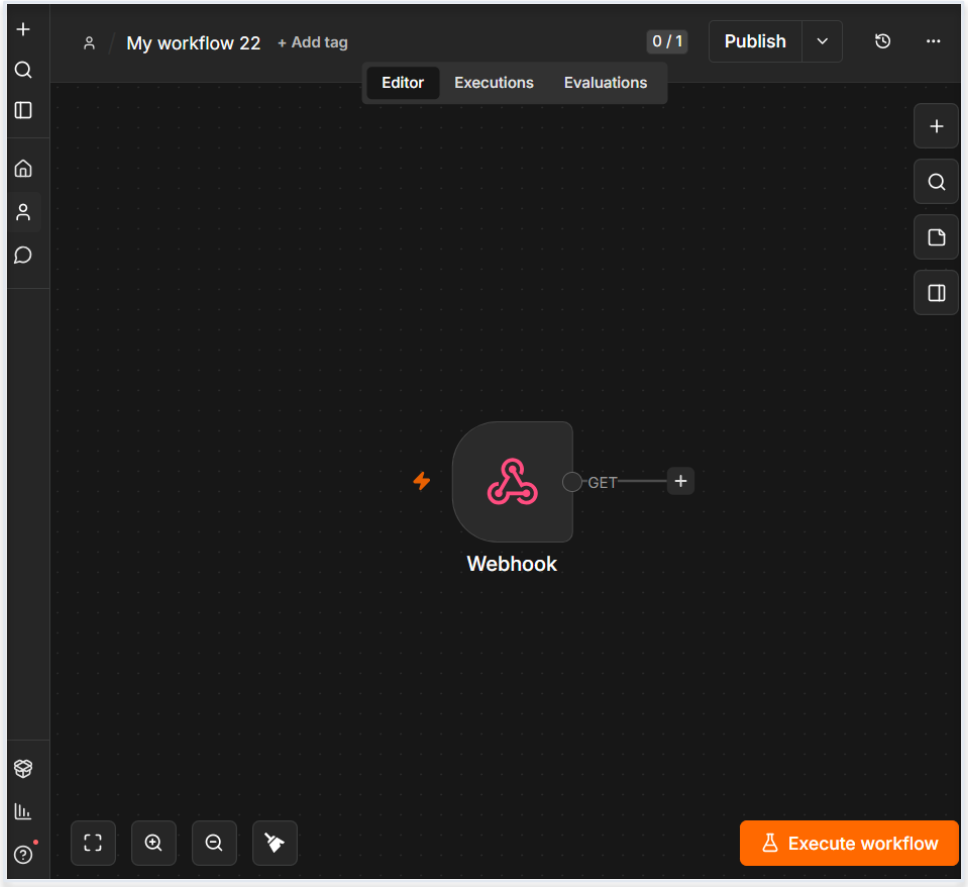
Sign In to n8n

1. Go to app.n8n.cloud/login — or open your own instance URL directly
2. Enter your assigned account email, e.g. n8n18oct5@tertiaryinfotech.com
3. Enter the password your trainer gives you
4. Open your instance from the dashboard — a free 14-day trial, active through the training
5. You land in n8n — open Assistant or Overview to start building

Your account is already set up — the name is your email prefix and onboarding is done. Do not change the password.

© 2026 Tertiary Infotech Academy Pte Ltd

The n8n Editor



n8n canvas: left sidebar navigation · centre canvas · node toolbar · Editor / Executions / Evaluations tabs

© 2026 Tertiary Infotech Academy Pte Ltd

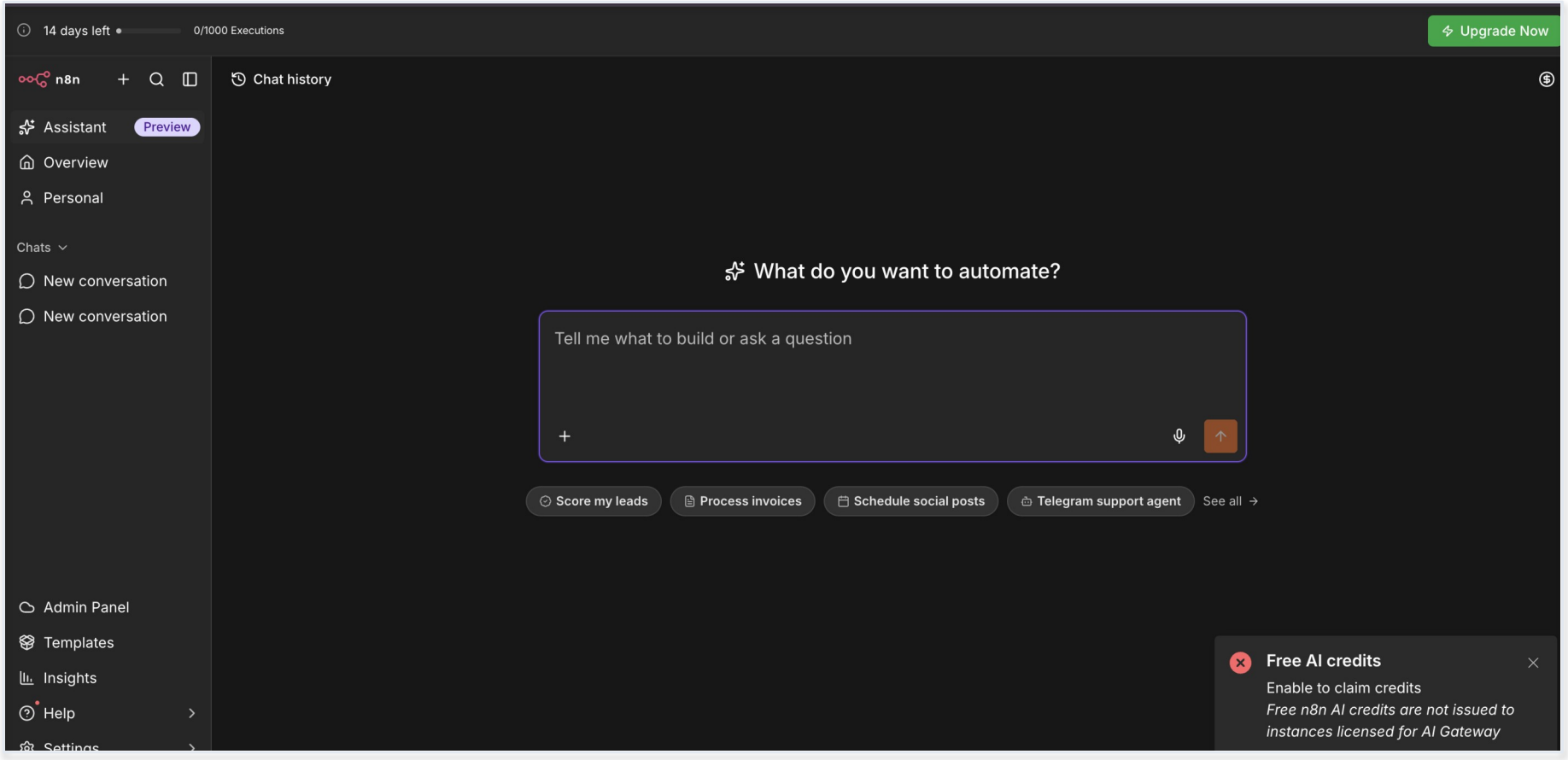
Claim Your Free AI Credits

1. Sign in to n8n Cloud with your training account — the free trial includes Gateway credits
2. Open a workflow and add an AI node, e.g. AI Agent → Gemini or OpenAI Chat Model
3. In the node's Credential field, choose Use Gateway credits — your balance shows beside it
4. Pick a supported model and run the node — no provider account or API key needed
5. Track the balance on the node, or in Admin Panel → Gateway credits

A red "Free AI credits — Enable to claim credits" toast is the legacy OpenAI offer: Gateway instances don't get it. Use Gateway credits.

© 2026 Tertiary Infotech Academy Pte Ltd

n8n Assistant — Describe It, n8n Builds It



Open Assistant in the sidebar and answer "What do you want to automate?" — it builds a normal workflow you can edit and publish.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · DEMO

Demo: n8n Assistant Builds a Social Media Campaign

Give n8n Assistant one clear prompt — it plans and builds the workflow below, with a human approval gate.

Form Trigger
you enter a topic

▶

Research
AI Agent + web search

▶

Draft post
AI writes the copy

▶

Human in the loop
email approve / reject

▶

Schedule post
wait, then publish

THE PROMPT

Create a workflow for a social media marketing campaign:
1. Start from a form where I enter a campaign topic.
2. Research the topic on the web and summarise 3 key points.
3. Draft a LinkedIn post under 150 words from the research.
4. Email me the draft and wait for my approval before posting.
5. If approved, schedule the post for 9am the next weekday;
   if rejected, log my feedback in a Google Sheet.

Use Gateway credits for the AI model. Ask me before publishing.

THEN YOU

- Review the plan before it builds
- Connect Gmail and the social account when it asks
- Test with a real topic and approve the draft
- Publish only when the run looks right

The AI built it; the workflow runs it — and nothing is posted until a person approves the draft.

© 2026 Tertiary Infotech Academy Pte Ltd

Trigger Nodes — What Is Allowed to Start a Workflow

The trigger you choose is a statement about who or what may start the process.

1 Manual Trigger

Click Execute Workflow — for building and testing, and for one-off jobs like the Activity 7b ingestion run.

2 Schedule Trigger

A clock starts the flow — cron or interval. Reports, reminders, nightly syncs.

3 Form Trigger

n8n hosts a web form for you; each submission starts a run. Activities 1–3.

4 Webhook

External systems call a URL you expose. Websites and apps — Activities 5, 7b, 8.

5 Chat / Telegram Trigger

Each chat message starts a run — n8n's chat UI or your Telegram bot. Activities 4, 6, 7a.

6 App & Email Triggers

On new Gmail, new Sheet row, new Drive file, new IMAP email — react to changes in apps you already use.

Pick the trigger by asking: who or what should be allowed to start this process?

© 2026 Tertiary Infotech Academy Pte Ltd

Triggers Used in This Course

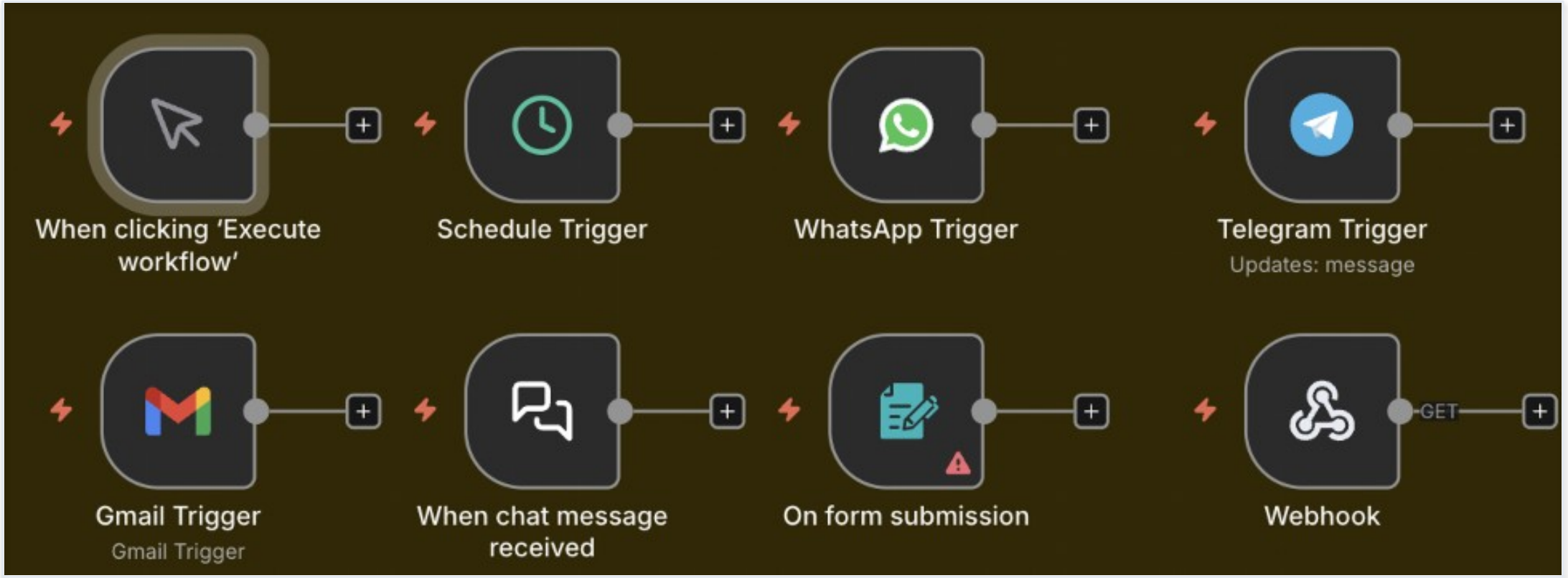
Every lab's trigger, and why that trigger is the right front door.

Trigger	Where	Why it fits
Form Trigger	Activities 1, 2, 3a, 3b	A visitor RSVP must start the flow
Telegram Trigger	Activities 4a, 4b, 6, 7a	Each bot message is one conversation turn
Webhook — POST	Activities 5, 7b, 8a, 8c	A website sends data and waits for a reply
Webhook — GET	Activity 8b	The dashboard pulls JSON from a URL
Manual Trigger	Activity 7b ingestion	A human runs the document load once

Schedule and app triggers follow the same pattern — swap the front door, keep the flow.

© 2026 Tertiary Infotech Academy Pte Ltd

Trigger Nodes on the n8n Canvas



n8n trigger nodes: Manual · Schedule · WhatsApp · Telegram · Gmail · Chat · Form · Webhook

© 2026 Tertiary Infotech Academy Pte Ltd

Data-Processing Nodes — The Workhorses

Between the trigger and the output, these nodes shape, route and transform the data.

1 Edit Fields (Set)

Add, rename or reshape fields — the node you reach for first when data is the wrong shape.

2 IF / Switch

Branch the flow: IF gives true/false paths; Switch gives one path per case.

3 Filter

Let only matching items continue — drop the rest of the list.

4 Merge

Combine two branches — append rows, match by key, or choose one side.

5 Split Out / Aggregate

Turn one item holding a list into many items — and back again.

6 Code

JavaScript or Python for the 5% a built-in node cannot do.

7 Loop / Wait

Process items in batches; pause a flow for time or an event.

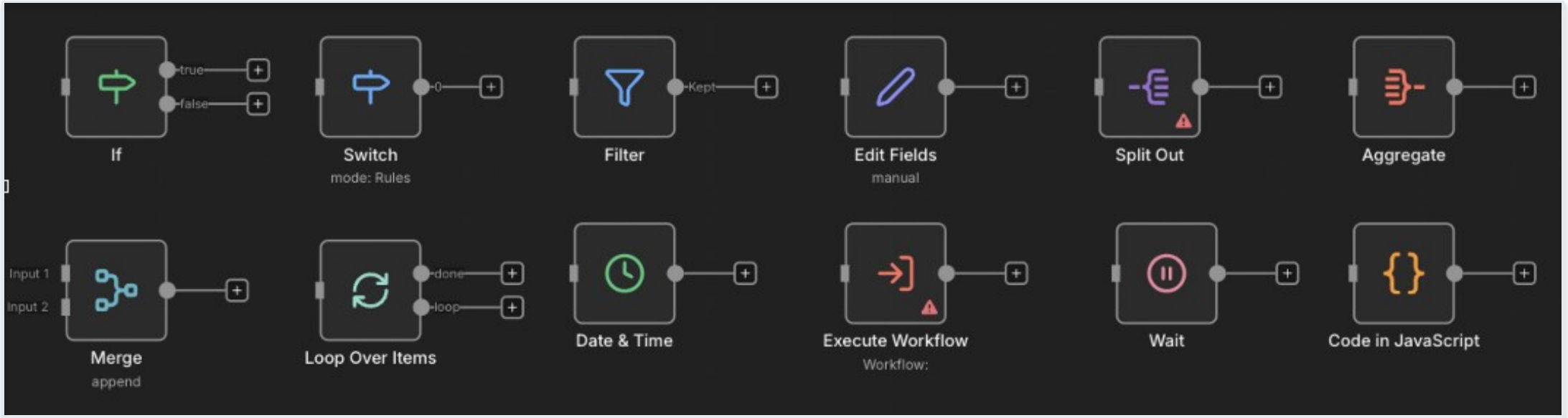
8 HTTP Request

Call any API on the internet — the universal escape hatch.

Most flows are: trigger → reshape (Set) → decide (IF) → act (app node). Reach for Code last.

© 2026 Tertiary Infotech Academy Pte Ltd

Data-Processing Nodes on the Canvas



If · Switch · Filter · Edit Fields · Split Out · Aggregate · Merge · Loop · Date & Time · Execute Workflow · Wait · Code

© 2026 Tertiary Infotech Academy Pte Ltd

Where the Data Goes — Storage & App Actions

Action nodes that write the result somewhere useful.

1 Data Table

n8n's built-in spreadsheet — instant, no credential. Trial Data Tables are NOT permanent; use for class demos.

2 Google Sheets / Excel

A real spreadsheet you keep — OAuth2 credential, Append Row. The persistent upgrade in Activity 3b.

3 Gmail / Outlook

Send (and in Activity 8a, Send and Wait for Response) — email as an output and as an approval gate.

4 Vector Store

Store embeddings for RAG — in-memory for demos, Pinecone / Supabase / Qdrant for production.

Anything you must keep goes to external storage — trial Data Tables vanish with the trial.

© 2026 Tertiary Infotech Academy Pte Ltd

JSON — How Data Travels Between Nodes

Every node passes a list of items; each item is a JSON object of key/value pairs.

- JSON (JavaScript Object Notation) is a simple text format for data.
- Values can be text, numbers, true/false, lists [ ] or nested objects { }.
- Most APIs send and receive JSON — so does every n8n node.
- Click any node to inspect its input and output JSON.

ONE FORM SUBMISSION AS JSON

```
{
  "Name": "Alice Tan",
  "Email": "alice@corp.com",
  "Attending": "Yes",
  "Pax": 2
}
```

Read any value later with an expression: {{ $json.Name }} → "Alice Tan"

© 2026 Tertiary Infotech Academy Pte Ltd

Expressions — The Token, Not the Text

An expression references a value that will exist at run time — it is resolved fresh on every run.

Trigger runs
a submission arrives

▶

Outputs exist
Name, Email, Attending…

▶

Later node references them
{{ $json.Name }}

▶

Value arrives at run time
different every run

✓ Mapped with the picker

Drag the field from the input panel — n8n writes the expression for you and shows a live preview of the resolved value.

✗ Typed as plain text

Type 'Name' as literal text and every email says 'Name'. If the path is wrong the value resolves to empty — not to an error.

Use expressions in any field — subject lines, URLs, conditions. Drag, don't type.

© 2026 Tertiary Infotech Academy Pte Ltd

Test vs Production Executions

The same workflow runs in two modes — know which one you are looking at.

TEST — in the editor

- Click Execute Workflow to run once.
- Data is shown live on the canvas, node by node.
- Pin a node's output to freeze it while you build downstream.
- Perfect for building — nothing needs to be Active.

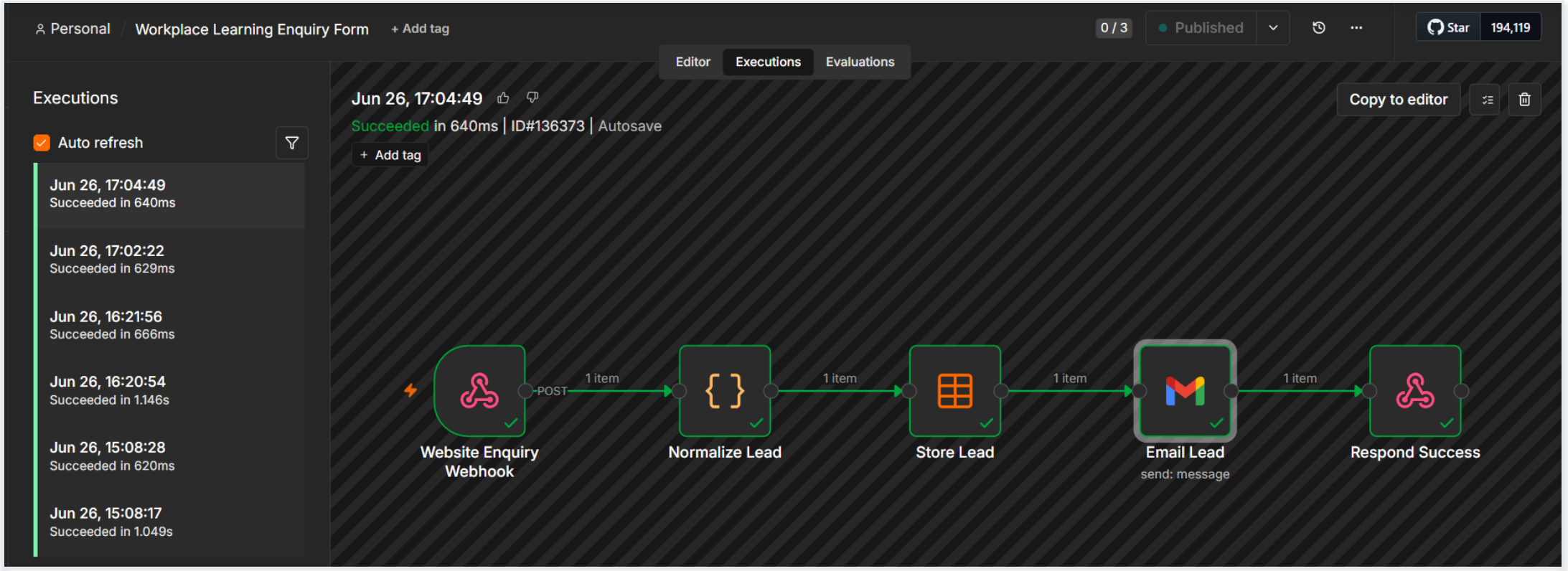
PRODUCTION — Active toggle ON

- Runs automatically when the trigger fires.
- Nothing appears on the canvas — runs are logged in Executions.
- Every run records status, timing and full data per node.
- This is what your users experience.

The Executions list is the honest account of what happened — debug from there, not from memory.

© 2026 Tertiary Infotech Academy Pte Ltd

Execution History



Executions tab — each run shows status, duration, ID and the full data at every node

© 2026 Tertiary Infotech Academy Pte Ltd

Credentials — Connect Your Accounts Once

Nodes reference credentials by name; set them up once under Credentials → Add credential.

1 Email

Gmail or Outlook via OAuth2 — used from Activity 1 onward.

2 LLM

Free n8n Gateway credits (no key needed), or your own OpenAI / Google Gemini API key — powers every AI agent from Activity 4.

3 Telegram

Bot token from @BotFather — the chat front door for Activities 4, 6, 7a.

4 Google Sheets

OAuth2 to your Google account — persistent storage in Activity 3b.

Imported workflows keep credential NAMES only — always re-select your own after importing.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · LAB WORKFLOW

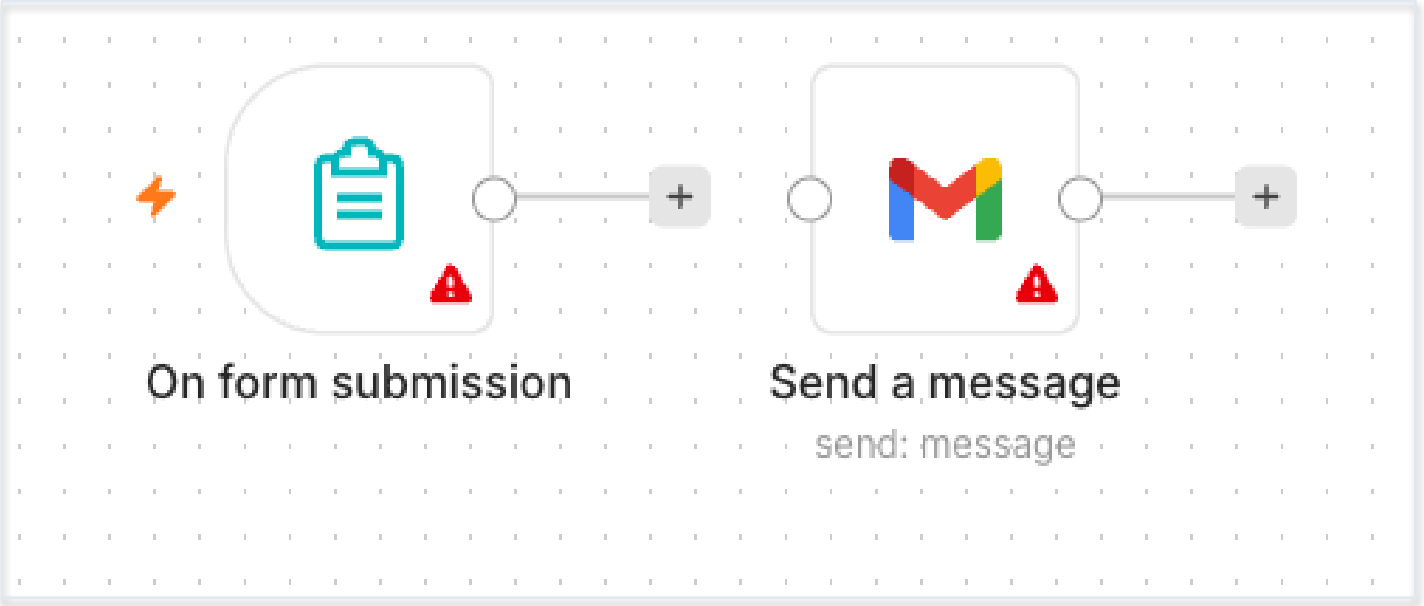
Activity 1 — Flyer with QR Code (Form → Email)

Collect a visitor's RSVP with a hosted n8n Form and email it to the admin — then put the form's QR code on a real event flyer.

Form Trigger
visitor submits the RSVP

▶

Gmail — Send
admin receives the details



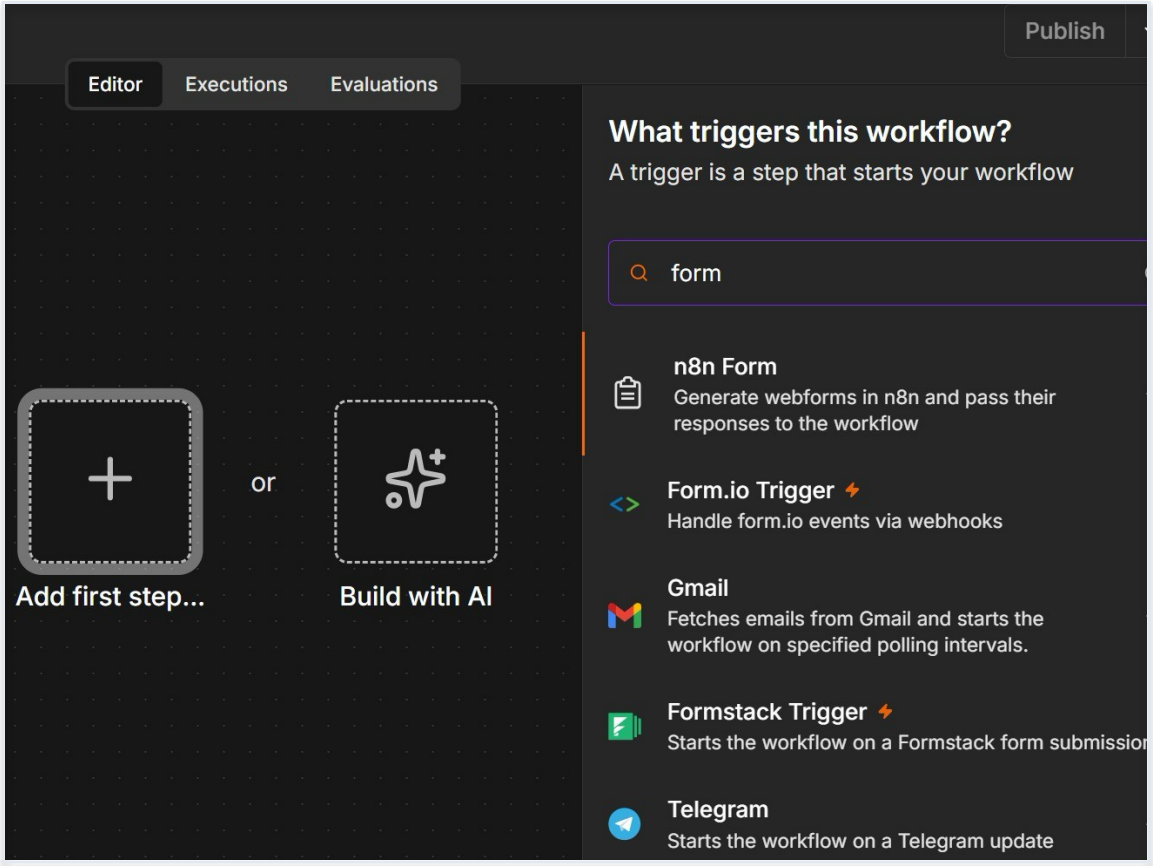
ACT 1

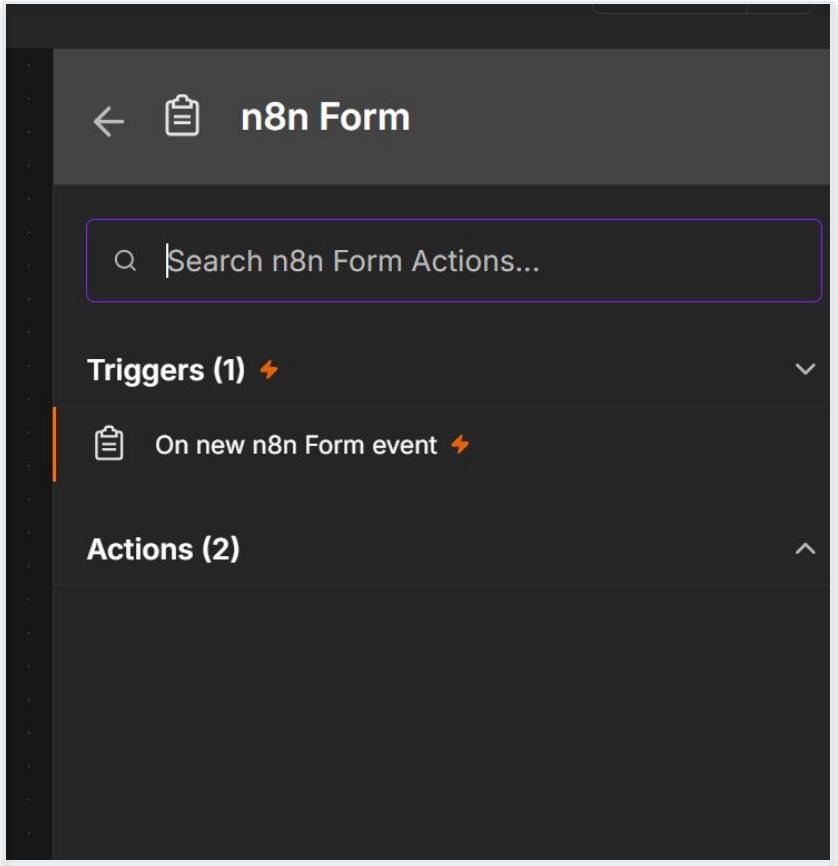
Flyer with QR Code (Form → Email)

Form Trigger → Gmail · Gmail OAuth2 credential · full steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

Activity 1 — Add the n8n Form Trigger



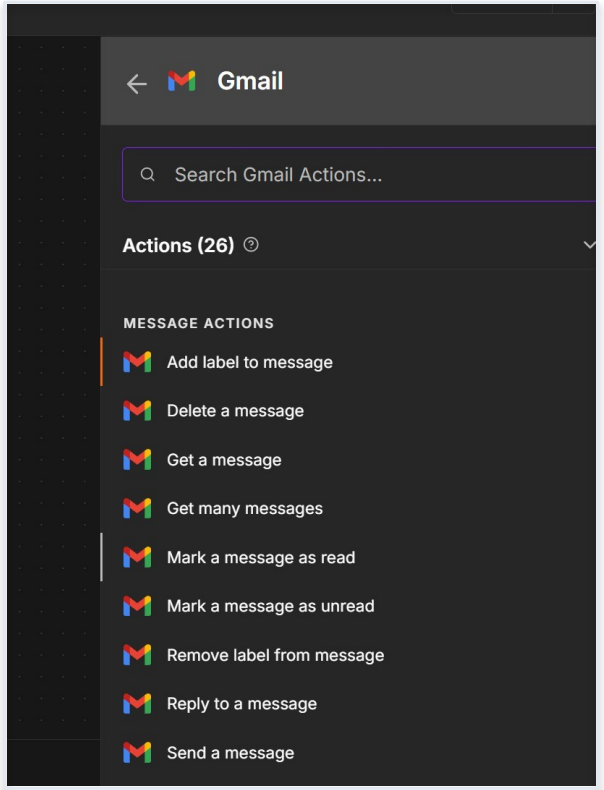


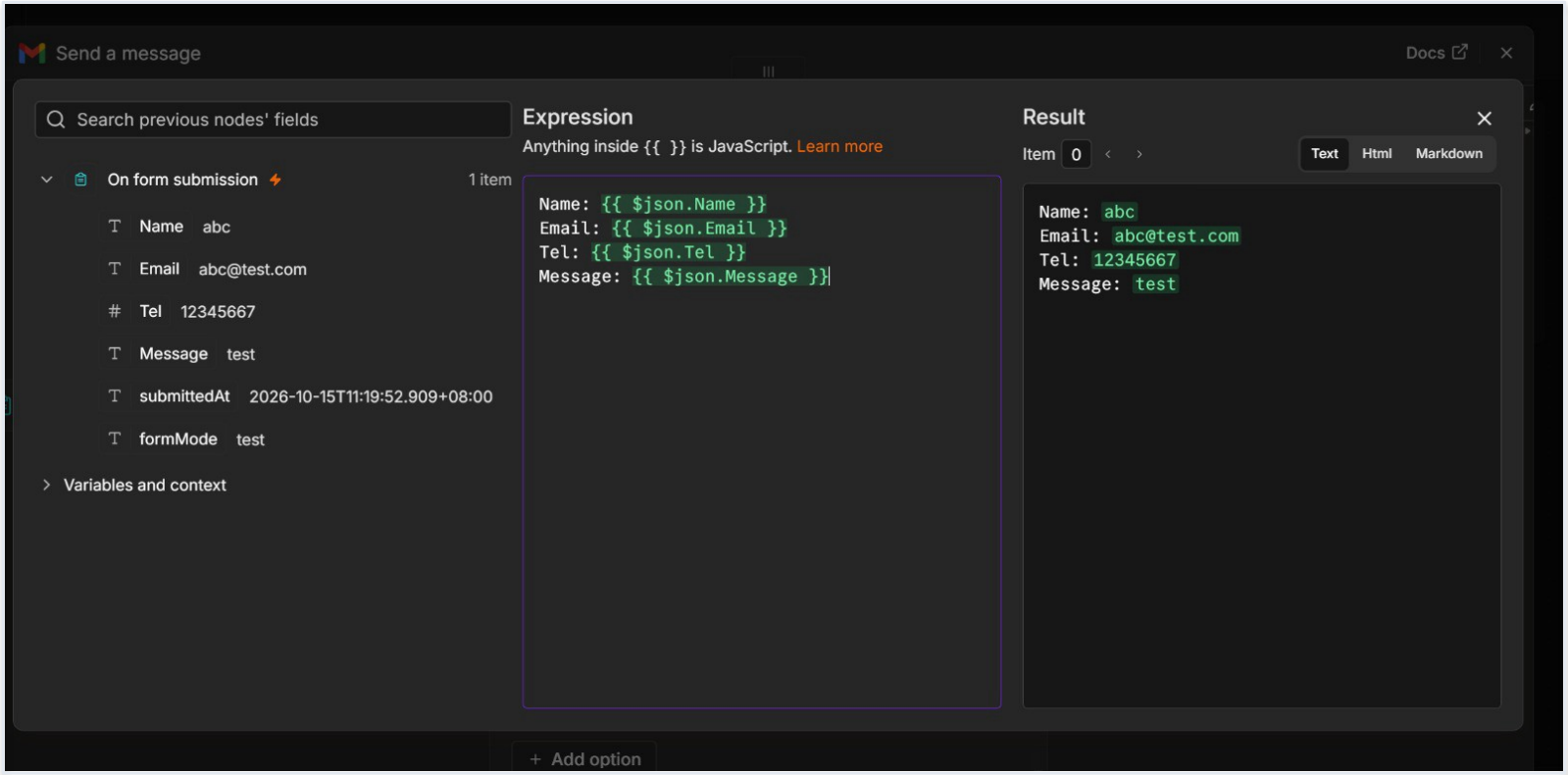
Click Add first step (+), search 'form' and choose n8n Form → On new n8n Form event. Then add the Name, Email, Phone and Message fields.

© 2026 Tertiary Infotech Academy Pte Ltd

ACT 1 · IN n8n

Activity 1 — Gmail: Send a Message with Expressions





Add Gmail → Send a message. Switch Message to Expression and drag in the form fields — the Result panel previews the email before you run it.

© 2026 Tertiary Infotech Academy Pte Ltd

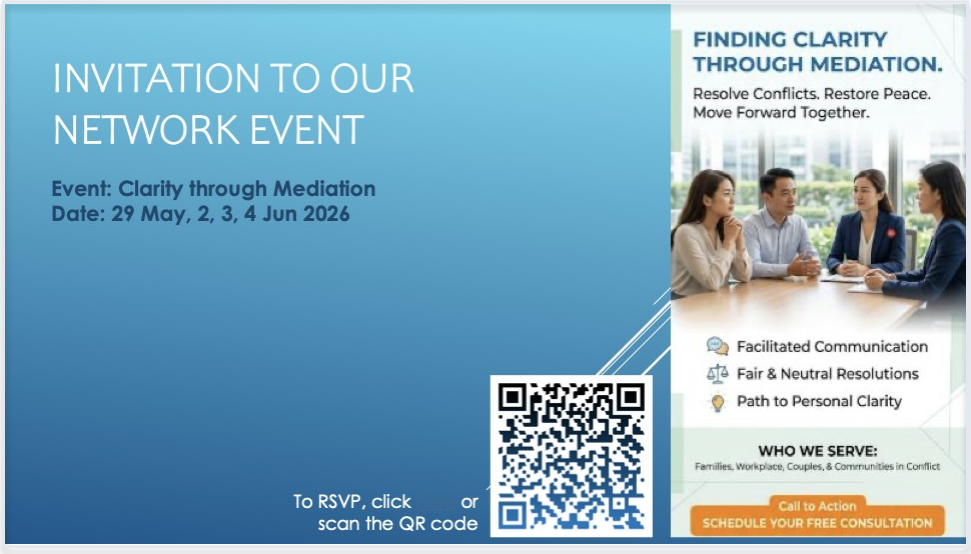
ACT 1 · GROUP ACTIVITY

Sample Flyers from Past Students









Each QR code encodes the form's Production URL — scan it, register, and the event administrator receives the details by email.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · LAB WORKFLOW

Activity 2 — Capture Submissions in a Data Table

Extend Activity 1 so every submission is also saved — your first taste of storing data, not just forwarding it.

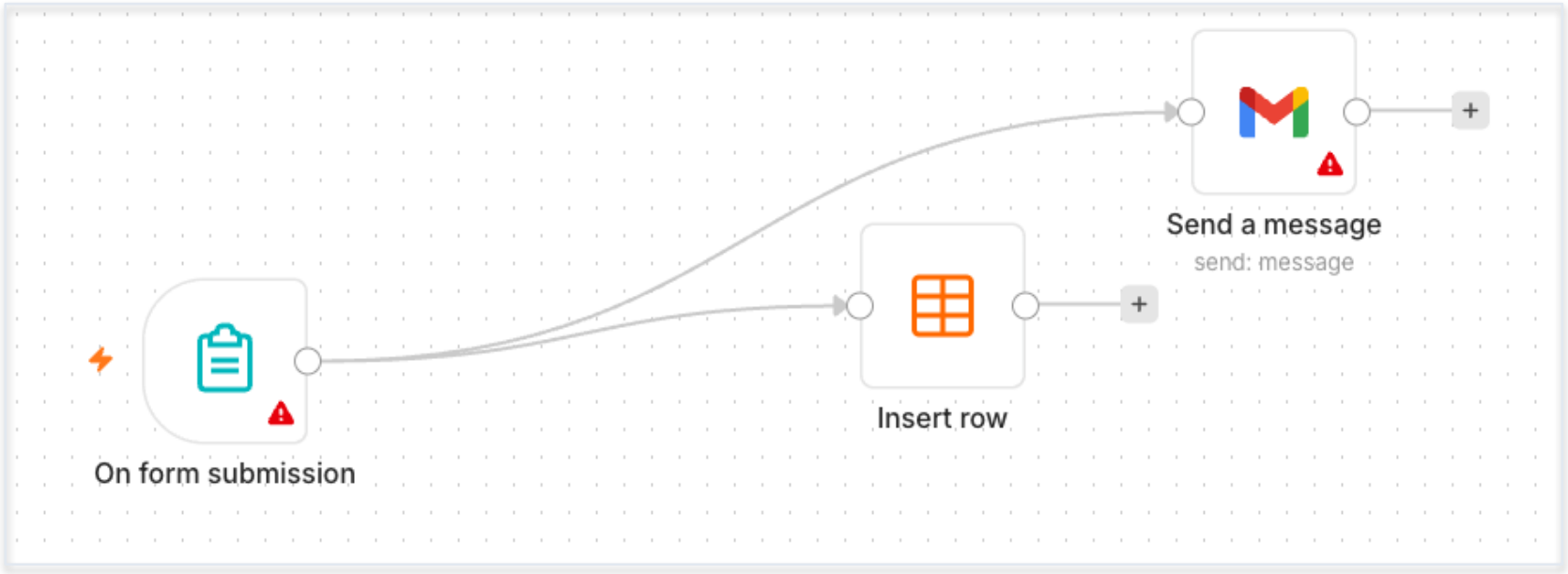
Form Trigger
same RSVP form

▶

Data Table — Insert Row
every submission saved

▶

Gmail — Send
admin still notified



ACT 2

Capture Submissions in a Data Table

Form Trigger → Data Table + Gmail · columns mapped with {{ $json.… }} expressions · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

Lunch Break

1 hour · see you at 2:00 pm

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · LAB WORKFLOW

Activity 3a — Conditional Response (Data Table)

Add decision-making: attending guests are saved; everyone else gets a polite thank-you email.

Form Trigger
+ Attending? Yes/No field

▶

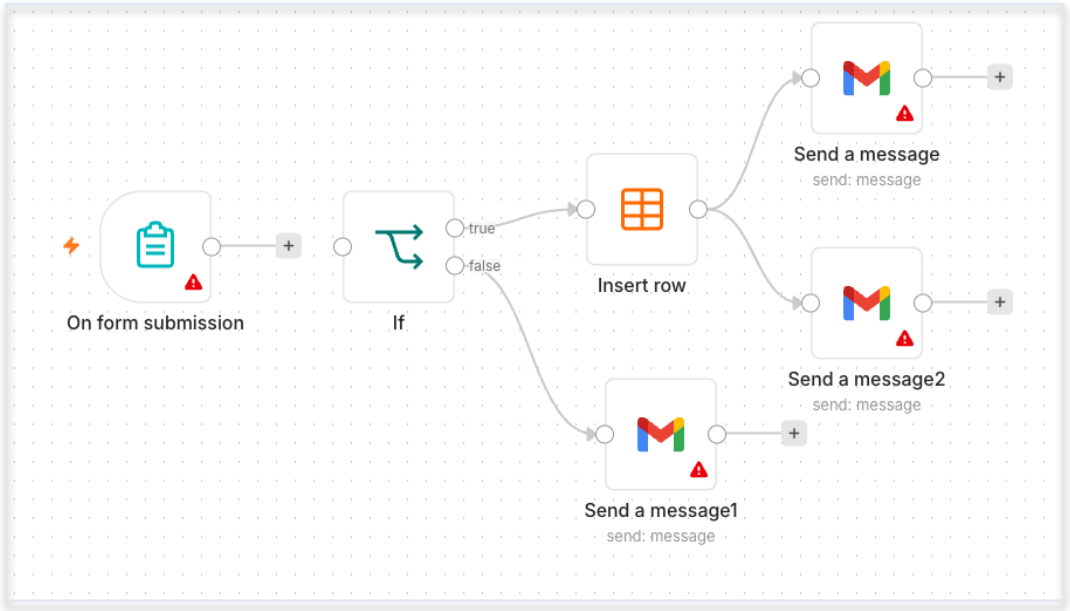
IF
Attending = Yes ?

▶

Data Table
YES → save the RSVP

▶

Gmail
NO → thank-you email



ACT 3a

Conditional Response (Data Table)

Form → IF → Data Table / Gmail · both branches must be built · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 1 · LAB WORKFLOW

Activity 3b — Conditional Response (Google Sheets)

Same logic, persistent storage: the YES branch appends to a real Google Sheet you keep after the trial.

Form Trigger
same form

▶

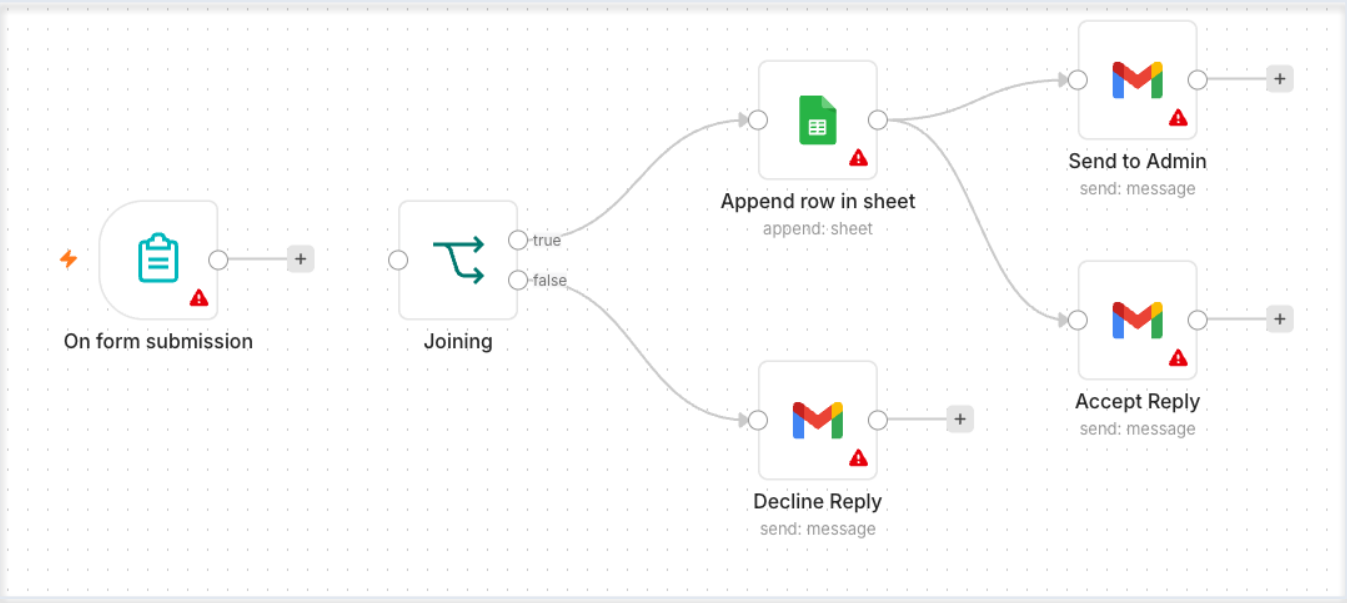
IF
Attending = Yes ?

▶

Google Sheets — Append
YES → a spreadsheet you keep

▶

Gmail
NO → thank-you email



ACT 3b

Conditional Response (Google Sheets)

Form → IF → Google Sheets / Gmail · Google Sheets OAuth2 · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

02

TOPIC 2

AI Agents

Components of an agent · model, memory, tools, instructions · model parameters · MCP

Applied in Activities 4a and 4b

© 2026 Tertiary Infotech Academy Pte Ltd

Components of an AI Agent

Four parts, one hub: the agent node wires them together — and only some are under your direct control.

MODEL

The LLM 'brain' — interprets language, reasons, chooses the next step. gpt-4.1-mini in this course.

MEMORY

Keeps the conversation — follow-up questions still make sense. Window buffer per chat/session.

THE AI AGENT

plans · decides · calls tools

TOOLS

Abilities it may use: Data Table lookups, vector stores, HTTP calls, sub-workflows.

INSTRUCTIONS

The system message — identity, scope, rules and refusals. Always in force.

Model + Memory + Tools + Instructions — every agent in this course is exactly these four parts.

© 2026 Tertiary Infotech Academy Pte Ltd

The Model — Popular LLM Providers

Any chat model plugs into the same AI Agent node — swap the brain, keep the agent.

Provider	Models	Notes
OpenAI	gpt-4.1, gpt-4.1-mini, gpt-4o	Course default: gpt-4.1-mini
Google	Gemini 2.x Flash / Pro	Used for embeddings in Activity 7
Anthropic	Claude (Opus, Sonnet, Haiku)	Strong reasoning and safety
Meta	Llama 3.x	Open weights — self-hostable
Mistral	Mistral / Mixtral	Open weights — EU option

Pick by task: small fast models for classification and guardrails, larger models for reasoning.

© 2026 Tertiary Infotech Academy Pte Ltd

Model Parameters — Temperature and Friends

Settings on the chat-model node shape how the model answers — the most important is temperature.

1 Temperature 0 – 0.3

Deterministic and repeatable. Use for classification, guardrails, data extraction — automation that must give the same decision twice.

2 Temperature ≈ 0.7

Balanced — natural-sounding chat with some variety. The default for customer-facing assistants.

3 Temperature 1.0 +

Creative and surprising. Marketing copy and brainstorming — never for decisions.

1 Top-p

Alternative sampling control — limits choices to the most probable tokens. Tune temperature OR top-p, not both.

2 Max tokens

Caps the length of the reply — protects cost and keeps chat answers short.

3 Frequency / presence penalty

Discourage repetition and nudge new topics — useful for long generations.

For business automation keep temperature LOW — the same input should produce the same decision.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · AI AGENTS

Memory — Why the Agent Remembers

Without memory every message is a stranger; with it, 'what about tomorrow?' makes sense.

Msg 1: 'Any leave left?'
agent answers: 12 days

▶

Stored in memory
keyed by chat / session ID

▶

Msg 2: 'What about Bob?'
agent knows the topic is leave

▶

Window slides
only the last N turns kept

Simple (buffer window) memory

Keeps the last N exchanges of THIS conversation — n8n's default, used in all course agents.

Session key

Memory is per chat ID — two Telegram users never see each other's context.

Memory is conversation context — it is NOT long-term knowledge. Facts live in tools and documents (RAG).

© 2026 Tertiary Infotech Academy Pte Ltd

Tools — What the Agent May Do

Attach tools to the agent's Tool input; the model decides when to call one, based on your description.

1 Data Table / Sheets tool

Look up structured records — the Employees table in Activity 4b.

2 Vector Store tool

Retrieve document chunks for grounded answers — the knowledge_base in Activity 7.

3 HTTP Request tool

Call any external API mid-conversation — prices, weather, CRM.

4 Sub-workflow tool

Wrap an entire n8n workflow as one callable tool — your own logic, on demand.

The tool DESCRIPTION is how the agent knows when to call it — write it for the model, not for a developer.

© 2026 Tertiary Infotech Academy Pte Ltd

Instructions — The Four Parts That Constrain

The system message is always in force. Every good agent instruction in this course states four things.

1 Identity

Who the agent is and what it must never do. 'You are HR Buddy, the HR assistant for staff of Acme Pte Ltd.'

2 Source rules

Where answers come from. 'Answer ONLY from the Employees tool / the uploaded documents.'

3 Refusals

What is out of bounds, stated positively. 'Do not reveal salaries or another person's records.'

4 Escalation

What to do at the boundary. 'If unsure, say so and direct the user to hr@acme.com.'

Vague instructions produce confident wrong answers — every rule you skip, the model improvises.

© 2026 Tertiary Infotech Academy Pte Ltd

Good vs Bad Instructions

The same bot, two system messages — one is testable, one is a hope.

✗ TOO VAGUE

"You are a helpful assistant. Answer the user's questions politely."

No identity · no sources · no refusals · no escalation.

It will happily invent leave balances and reveal other people's data — politely.

✓ CONSTRAINED

"You are HR Buddy, the HR assistant for Acme staff.

Answer only questions on leave policy, payroll dates and claims.

Use the Employees tool for any staff lookup — never answer from general knowledge.

Never reveal salaries or another employee's personal data.

If you are not sure, say so and direct the user to hr@acme.com."

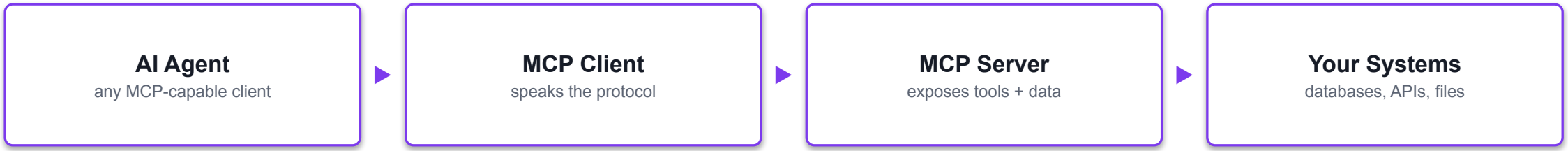
Identity · source rules · refusals · escalation — you can test each line by trying to break it.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · AI AGENTS

MCP — Model Context Protocol

An open standard that gives agents a common plug for tools and data — instead of one custom integration per pair.



Without MCP

Every agent × every tool needs its own custom integration — N agents and M tools means N × M connectors to build and maintain.

With MCP

Build ONE MCP server per system; any MCP-capable agent can use it. N + M pieces, and tools become reusable across agents.

MCP is like USB-C for AI tools — one connector, many devices.

© 2026 Tertiary Infotech Academy Pte Ltd

MCP in n8n

MCP SERVER TRIGGER — n8n serves tools

- Expose your n8n workflows AS tools over MCP.
- Claude, IDEs or other agents call your flows directly.
- n8n becomes the tool layer for any MCP client.
- Auth via bearer token on the server URL.

MCP CLIENT TOOL — n8n consumes tools

- Attach external MCP servers to your n8n AI Agent.
- The agent gains their tools without custom nodes.
- Mix MCP tools with native n8n tools freely.
- Same agent loop — more reach.

Two directions, one protocol: n8n can serve tools to other agents, and borrow tools for its own.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · LAB WORKFLOW

Activity 4a — An HR Policy Chatbot on Telegram

What is a chatbot? An agent behind a chat channel. Your first agent, Ava, answers staff questions on company HR policies.

Telegram Trigger
an employee messages the bot

▶

HR Agent — Ava
HR policies + memory

▶

Chat model
Gateway credits, no API key

▶

Reply on Telegram
back to the same chat

1 What Ava knows

Sample HR policies written into the system instruction — leave, sick leave, claims, hours, probation.

2 Guardrails in the prompt

Answer only from the policies, never invent a number, and refer salary, medical or disciplinary questions to HR.

3 Then give her a tool

Activity 4b attaches an Employees Data Table, so Ava can also answer from real staff records.

You will: create a bot with BotFather → build the workflow with n8n Assistant → connect, publish and chat.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · ACTIVITY 4a

Step 1: Create a Telegram Bot with BotFather

1. Open Telegram and search for BotFather (blue checkmark)
2. Start a chat and send /newbot
3. Type a display name, e.g. My Free Helper Bot
4. Choose a unique username ending in bot, e.g. my_free_helper_bot
5. Copy the HTTP API token BotFather sends back

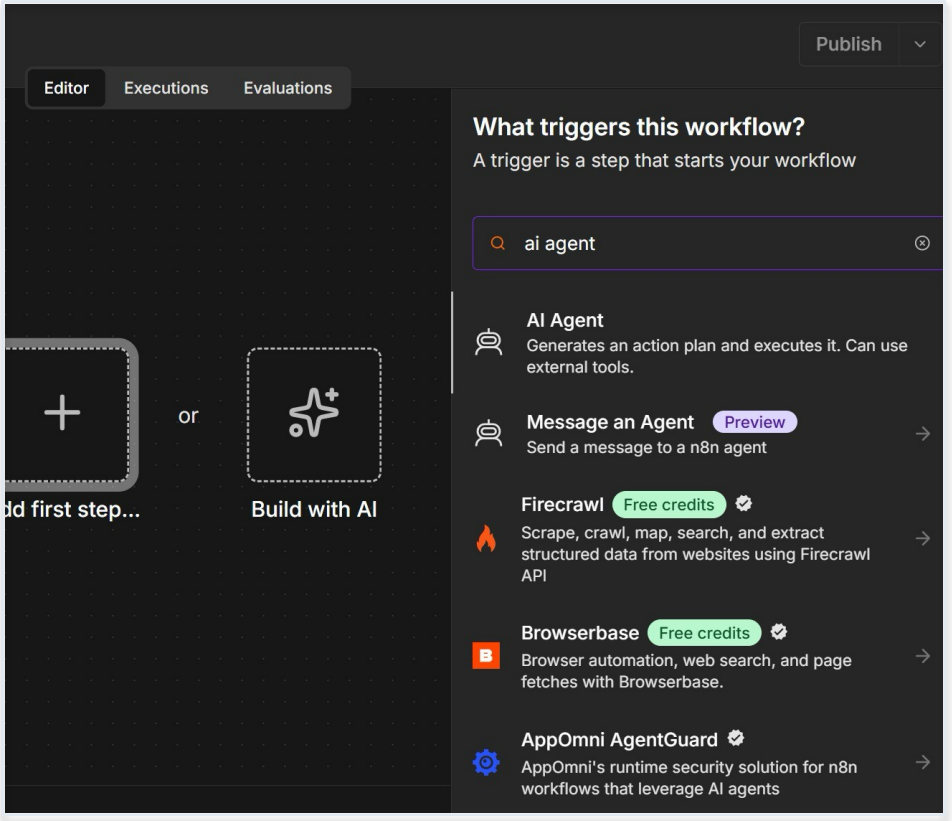
The token is the bot's password — never share or commit it. If it leaks, send /revoke to BotFather for a new one.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · ACTIVITY 4a

Step 2a: Open n8n Assistant with Build with AI

- Create a new workflow. The empty canvas offers two starts: Add first step, or Build with AI.
- Click Build with AI — this opens n8n Assistant, which builds the workflow from a plain-English prompt.
- Add first step is the manual route: it opens the node search, here filtered to 'ai agent'.
- Either way you get the same nodes. The Assistant just writes the first draft for you.



Build with AI drafts the workflow; you still review every node before you publish.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · ACTIVITY 4a

Step 2b: Paste the Prompt into n8n Assistant

Create a new workflow, open n8n Assistant and paste this prompt — it plans and builds the chatbot for you.

THE PROMPT

Create a 3-node workflow:

- A user posts a message to a Telegram bot.
- Telegram sends the data to an HR agent.
- The HR agent responds to the user on the company HR policies. Create a system instruction for the HR agent with sample company policies in it. The HR agent has a memory.
- Respond to Telegram.

THEN YOU

- Review the plan before it builds
- Read the system instruction it wrote — edit the sample policies if you like
- Check a memory node is attached to the HR agent
- Stuck? Import the reference solution instead

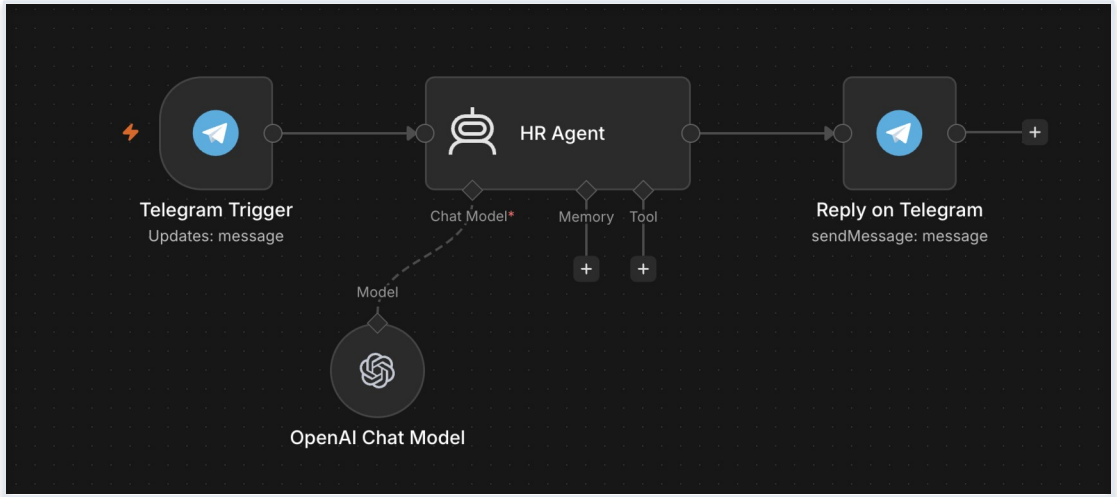
Reference solution: labs/activity4-telegram-agent/Activity4a-Telegram-HR-Agent.json

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · ACTIVITY 4a

The Workflow n8n Assistant Builds

Your canvas should look like this: trigger, agent, reply — with the chat model and memory hanging under the agent.



1 Telegram Trigger

Updates: message — every message sent to the bot starts one execution.

2 HR Agent

Ava's role, rules and sample HR policies live in the system instruction n8n Assistant wrote.

3 Model and memory

Chat model on Gateway credits; memory keeps each employee's conversation, keyed on the chat ID.

4 Reply out

Reply on Telegram sends back to the chat ID the trigger received, so the reply lands in the same chat.

Shown before memory was added — check yours has a memory node (the reference file has one).

© 2026 Tertiary Infotech Academy Pte Ltd

Step 3: Connect the Bot, Publish and Chat

1. Telegram Trigger → Credential → Create new → paste the token
2. Telegram reply node → select the same credential
3. Chat model → Use Gateway credits
4. Publish — n8n registers its webhook with Telegram
5. Open t.me/<username>, tap Start, ask about annual leave

A Telegram bot has ONE webhook at a time — testing in the editor swaps in the Test URL, so publish again when you are done.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · ACTIVITY 4a

Chatting with Ava on Telegram

- /start opens the chat — Ava introduces herself exactly as her system message tells her to.
- Ask how many days of leave you get: the answer quotes the policy — 14, 18 or 21 days by length of service.
- Ask a follow-up — memory keeps the context. Then ask about a colleague's salary: Ava must refer you to HR.
- Every message is one workflow run — open Executions in n8n to watch it happen.



The bot is only as safe as its system message. Activity 4b gives Ava a tool: the Employees Data Table.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 2 · LAB WORKFLOW

Activity 4b — Agent + Data Table Tool (HR Admin)

Give Ava a tool: an Employees Data Table (100 records) she can query — answers now come from real staff records.

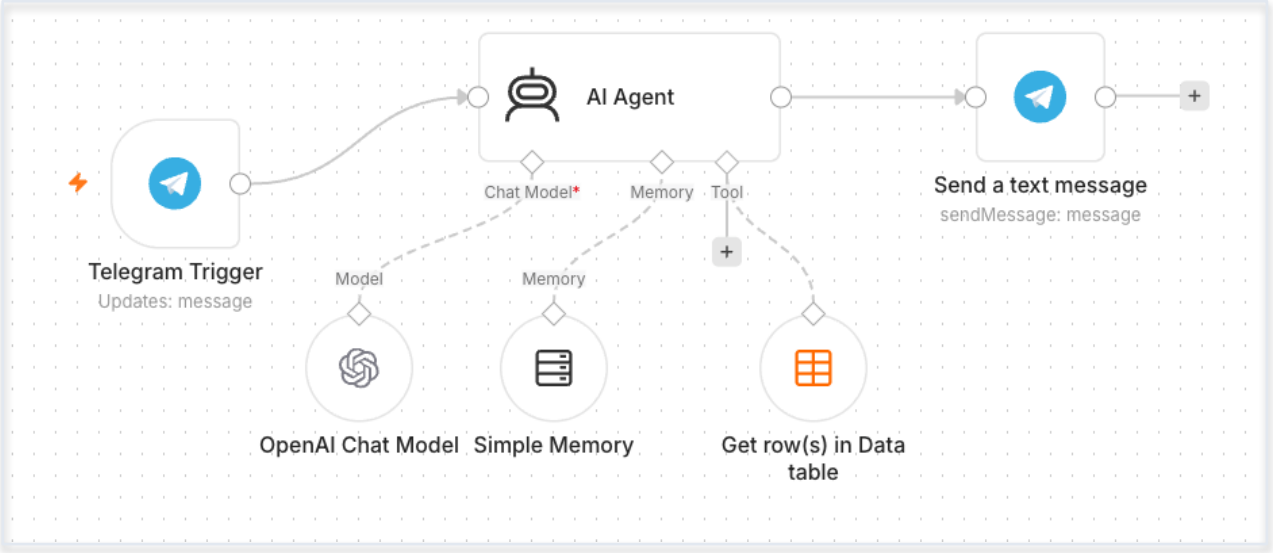
Telegram Trigger
same bot

▶

AI Agent
+ Employees Data Table tool

▶

Telegram — Send
answers grounded in the table



Reference layout — in your build the agent is Ava (HR Agent) and the reply node is Reply on Telegram.

ACT 4b

Agent + Data Table Tool (HR Admin)

Tool description tells the agent WHEN to use it · instruction updated to route staff questions · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

WRAP-UP

End of Day 1 — Recap

- You built form automations, conditional logic and storage (Data Table → Google Sheets).
- You met the agent's four parts — model, memory, tools, instructions — and MCP.
- You shipped a Telegram AI agent, then grounded it in an Employees table.
- Tomorrow: webhooks, external APIs, and RAG.

© 2026 Tertiary Infotech Academy Pte Ltd

03

TOPIC 3

Webhooks

The inbound URL · GET vs POST · Test vs Production URL · responding to the caller

Applied in Activity 5

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 3 · WEBHOOKS

What Is a Webhook?

A URL your workflow listens on. When the outside world calls it, your workflow runs — the reverse of you calling an API.

Event happens
visitor sends a chat message

▶

HTTP request to your URL
the website POSTs JSON

▶

Workflow runs
n8n receives the data as input

▶

Respond to Webhook
your reply returns to the caller

1 Push, not pull

No polling, no schedule — the caller notifies you the moment something happens.

2 Any caller

Websites, payment providers, forms, apps — anything that can send an HTTP request.

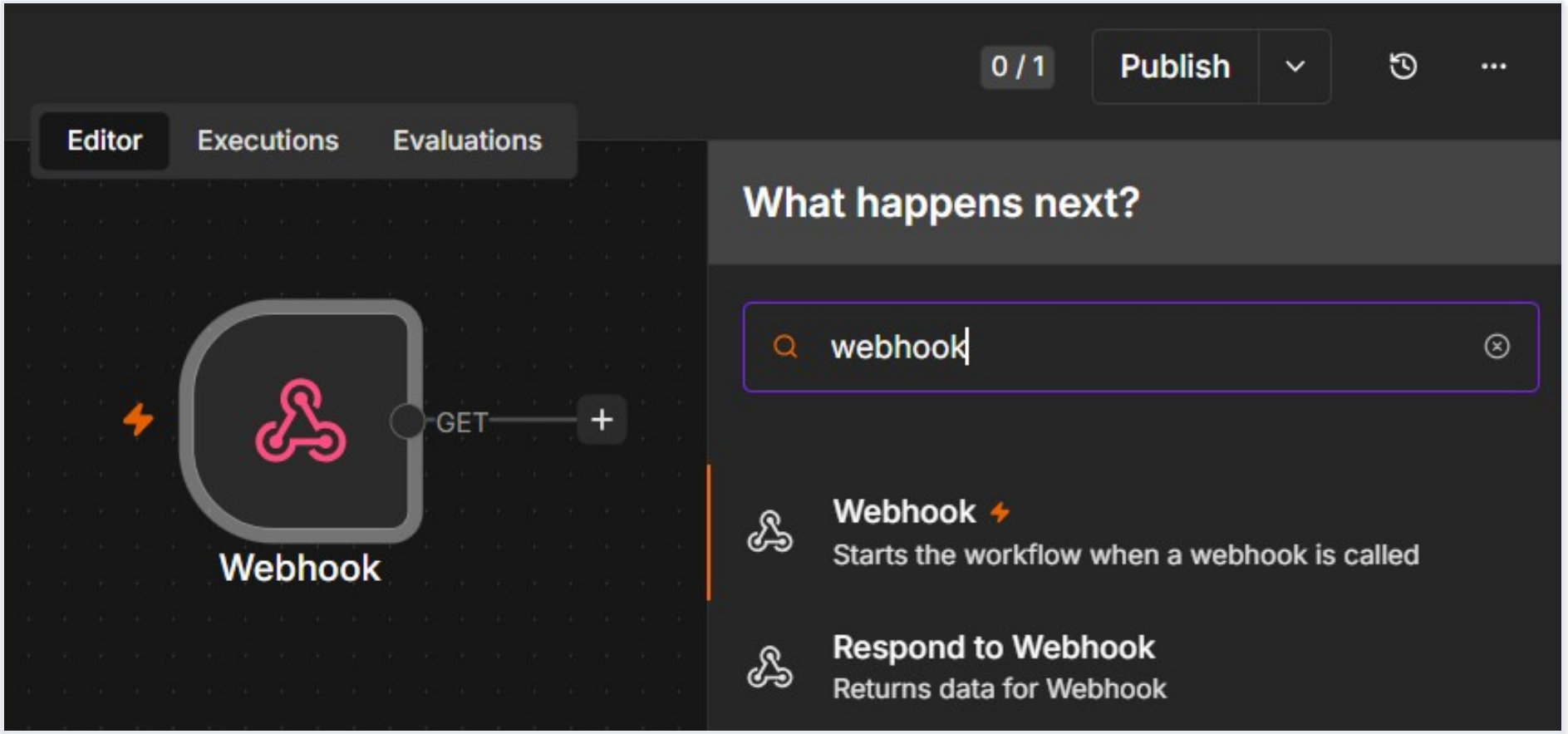
3 Paired node

Webhook starts the flow; Respond to Webhook sends the answer back on the same connection.

A webhook is the reverse of an API call — the outside world calls YOU when an event happens.

© 2026 Tertiary Infotech Academy Pte Ltd

Webhook Nodes in n8n



Webhook node starts the workflow when called · Respond to Webhook node returns data to the caller

© 2026 Tertiary Infotech Academy Pte Ltd

Webhook vs HTTP Request — Direction of the Call

WEBHOOK — inbound

- An external service calls YOUR workflow.
- You react the moment an event happens.
- The trigger of the flow.
- Activity 5: the website calls your agent.

HTTP REQUEST — outbound

- YOUR workflow calls an external service.
- You pull data when you choose.
- An action inside the flow.
- Activity 6: your flow calls Twelve Data.

Same HTTP underneath — the only difference is who dials.

© 2026 Tertiary Infotech Academy Pte Ltd

HTTP Methods on a Webhook — GET vs POST

The method is part of the contract with the caller: where the data rides, and what the call means.

GET — read

- Data rides in the URL query string:
  - ?email=alice@corp.com
- A browser can call it by just visiting the URL.
- Read in n8n as {{ $json.query.email }}.
- Activity 8b: the dashboard GETs its leave data.

POST — send

- Data rides in the request body as JSON:
  - { "message": "How much is the course?" }
- Used by website chat, forms and app callbacks.
- Read in n8n as {{ $json.body.message }}.
- Activities 5, 7b, 8a, 8c: the site POSTs to the flow.

GET reads, POST sends. Set the method on the Webhook node — the caller must match it.

© 2026 Tertiary Infotech Academy Pte Ltd

Test URL vs Production URL

Every Webhook node gives you TWO URLs — they behave differently, and mixing them up is the #1 webhook bug.

TEST URL — /webhook-test/…

- Works only while you click 'Listen for test event'.
- Handles ONE call, then stops listening.
- The run appears live on the canvas — great for debugging.
- Dies the moment you leave the editor.

PRODUCTION URL — /webhook/…

- Requires the workflow toggled Active.
- Always on — every call starts a run.
- Runs are logged in the Executions list, not on the canvas.
- This is the URL you give to websites and apps.

Websites must call the PRODUCTION URL of an ACTIVE workflow — the Test URL stops when you leave the editor.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 3 · WEBHOOKS

Configuring the Webhook Node

Four settings decide whether the browser's call succeeds.

1 Method + Path

GET or POST, and the path that ends your URL — e.g. POST /webhook/hr-chat.

2 Allowed Origins (CORS)

Browsers block cross-site calls unless allowed — set * in class so a local HTML page can call n8n.

3 Response mode

Immediately (ack), when the flow finishes, or via a Respond to Webhook node — the course flows use Respond to Webhook.

4 Authentication

None for demos; Basic / Header / JWT for anything real — a public URL is a public door.

'Failed to fetch' in the browser is usually CORS or a dead Test URL — check both first.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 3 · LAB WORKFLOW

Activity 5 — Website Chatbot via Webhook (Investment Advisor)

Put your agent behind a public web page: the enquiry form and the floating chatbot both POST to one n8n webhook.

Webhook — POST
chat + enquiry form call it

▶

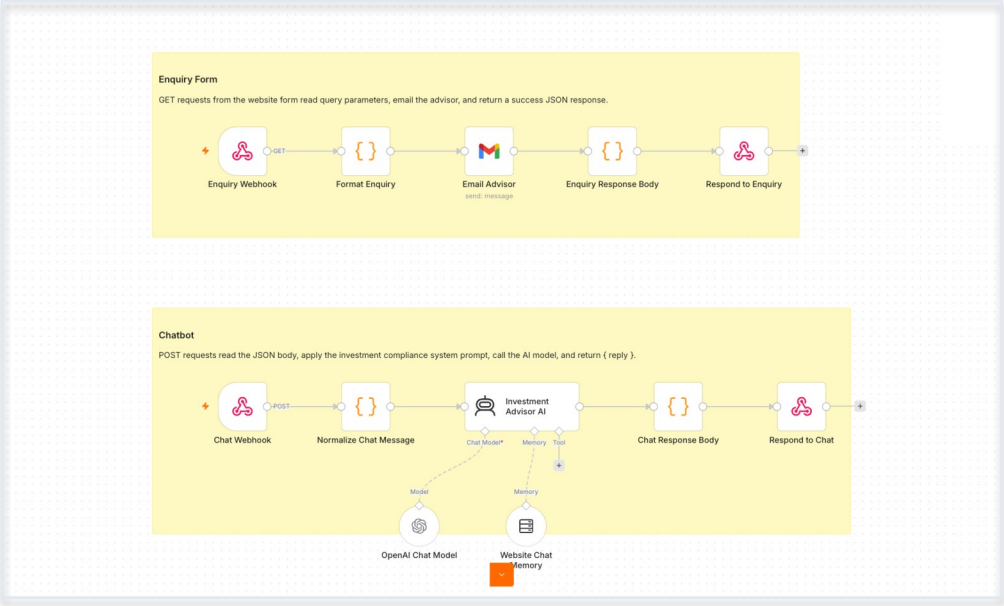
AI Agent
compliance instruction — no advice

▶

Respond to Webhook
reply returns to the page

▶

Gmail
enquiry → advisor inbox



ACT 5

Website Chatbot via Webhook (Investment Advisor)

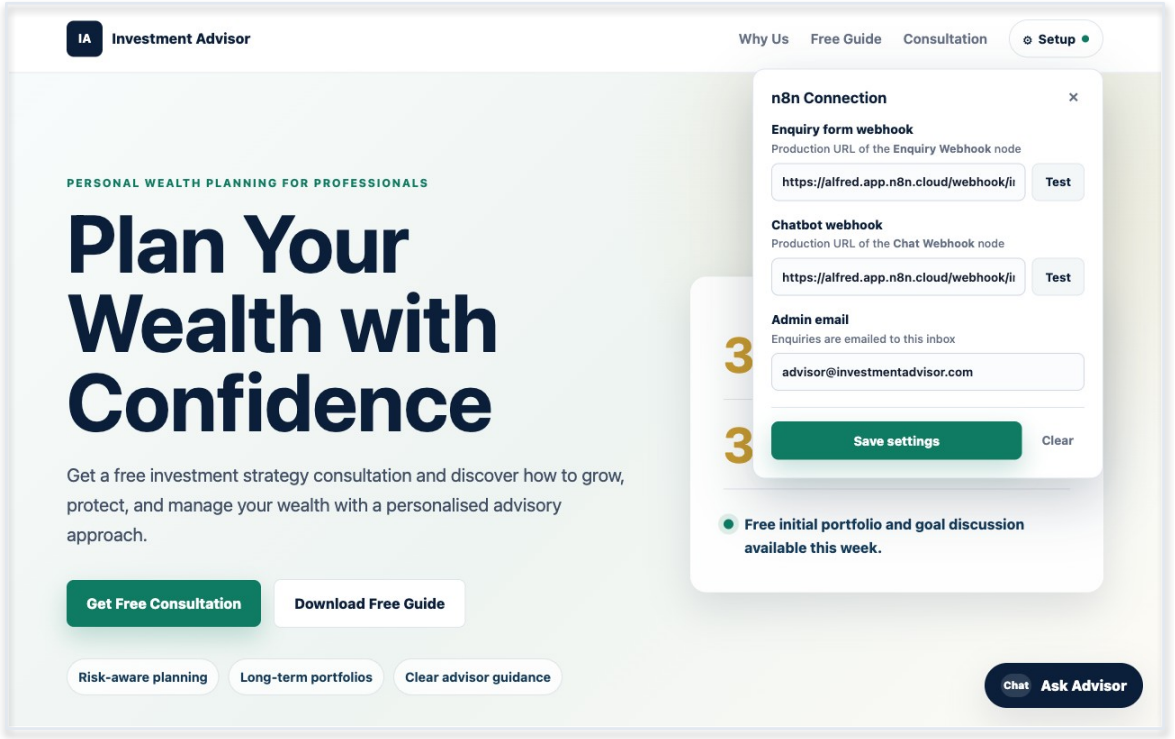
Production URL pasted into script.js · CORS = * · workflow Active · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

ACT 5 · THE WEBSITE

Activity 5 — The Investment Advisor Website

- A public landing page with an enquiry form and a floating 'Ask Advisor' chatbot.
- Both the form and the chat POST to one n8n Webhook.
- The AI Agent replies in the chat; the advisor is emailed the enquiry.
- Set the webhook Production URL in script.js, then open index.html.



© 2026 Tertiary Infotech Academy Pte Ltd

Tea Break

15 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

04

TOPIC 4

APIs and the HTTP Request Node

Calling external services · request anatomy · status codes · API keys

Applied in Activity 6

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 4 · API & HTTP

What Is an API?

A service's front desk: you send a structured request over HTTP, it returns structured data — usually JSON.

Your workflow
the client

▶

HTTP request
method + URL + key + params

▶

API server
processes the request

▶

JSON response
fields your next node can read

1 Live data on demand

Prices, weather, news, CRM records — data your workflow pulls the moment it needs it.

2 You dial out

The opposite of a webhook: YOUR flow calls THEIR server, on your schedule.

3 One node fits all

The HTTP Request node calls any API on the internet — even ones n8n has no integration for.

If a service has an API, n8n can use it — the HTTP Request node is the universal connector.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 4 · API & HTTP

Anatomy of an HTTP Request

Four parts you configure on the HTTP Request node.

Part	What it is	Activity 6 example
Method	GET (read) · POST (send) · PUT · DELETE	GET — read market candles
URL / endpoint	The address of the resource	api.twelvedata.com/time_series
Headers	Metadata, incl. authentication	(Twelve Data uses a query key instead)
Query params / body	The inputs you send	symbol=AAPL & interval=1min & apikey=…

Store API keys in Credentials — never hard-code them into the URL you share.

© 2026 Tertiary Infotech Academy Pte Ltd

Reading the Response — Status Codes

The status code tells you whose problem it is before you read a single byte of the body.

1 2xx — success

200 OK, 201 Created. The JSON body holds your data — map it to the next node.

2 401 / 403 — auth

Bad or missing API key. Check the credential, not the flow.

3 429 — rate limited

Too many calls for your plan — slow down or upgrade. Free tiers hit this fast.

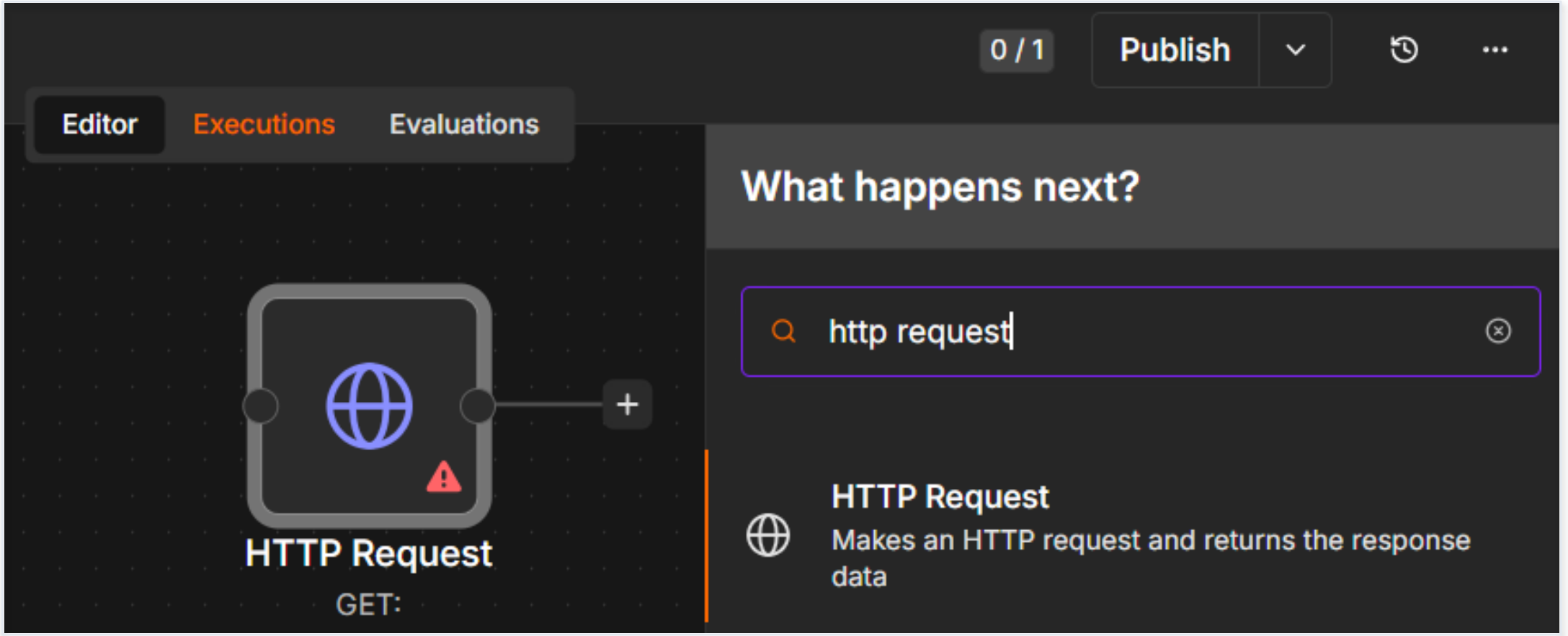
4 4xx vs 5xx

4xx: your request is wrong. 5xx: their server is broken. Retry helps only with 5xx.

4xx = fix your request · 5xx = wait and retry — never guess from the body alone.

© 2026 Tertiary Infotech Academy Pte Ltd

The HTTP Request Node

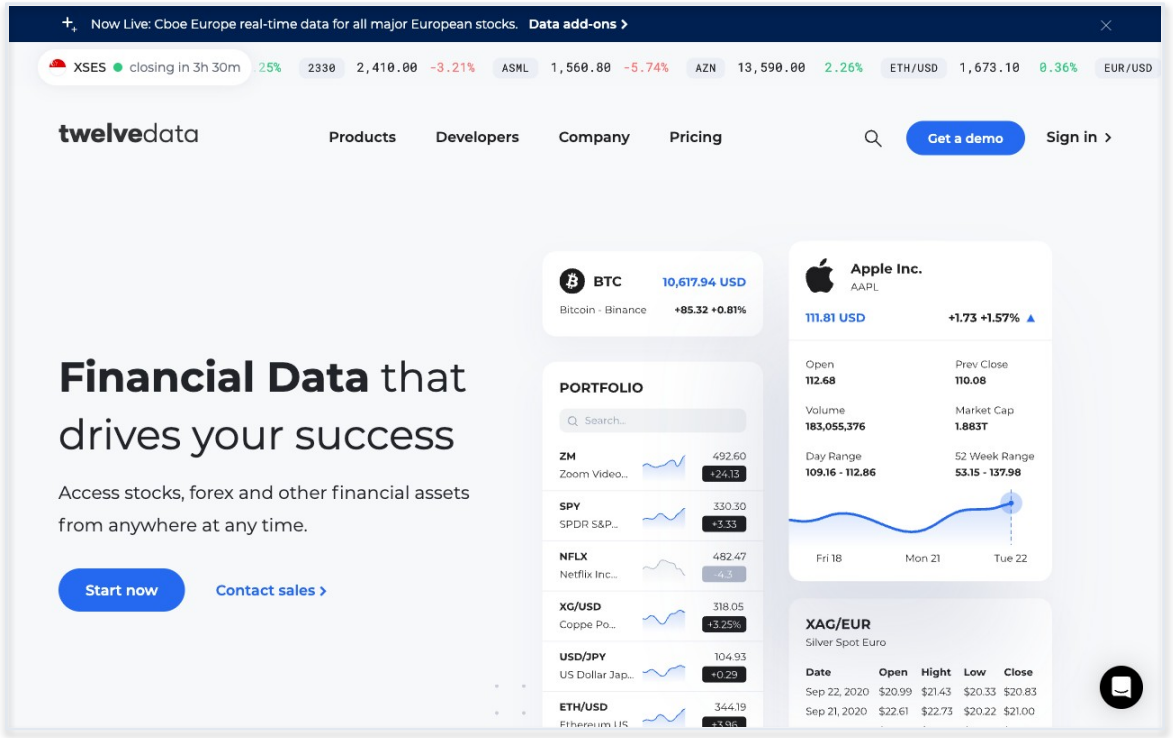


HTTP Request node — configure method, URL, auth, query parameters; the JSON response flows to the next node

© 2026 Tertiary Infotech Academy Pte Ltd

Twelve Data — Market Data API

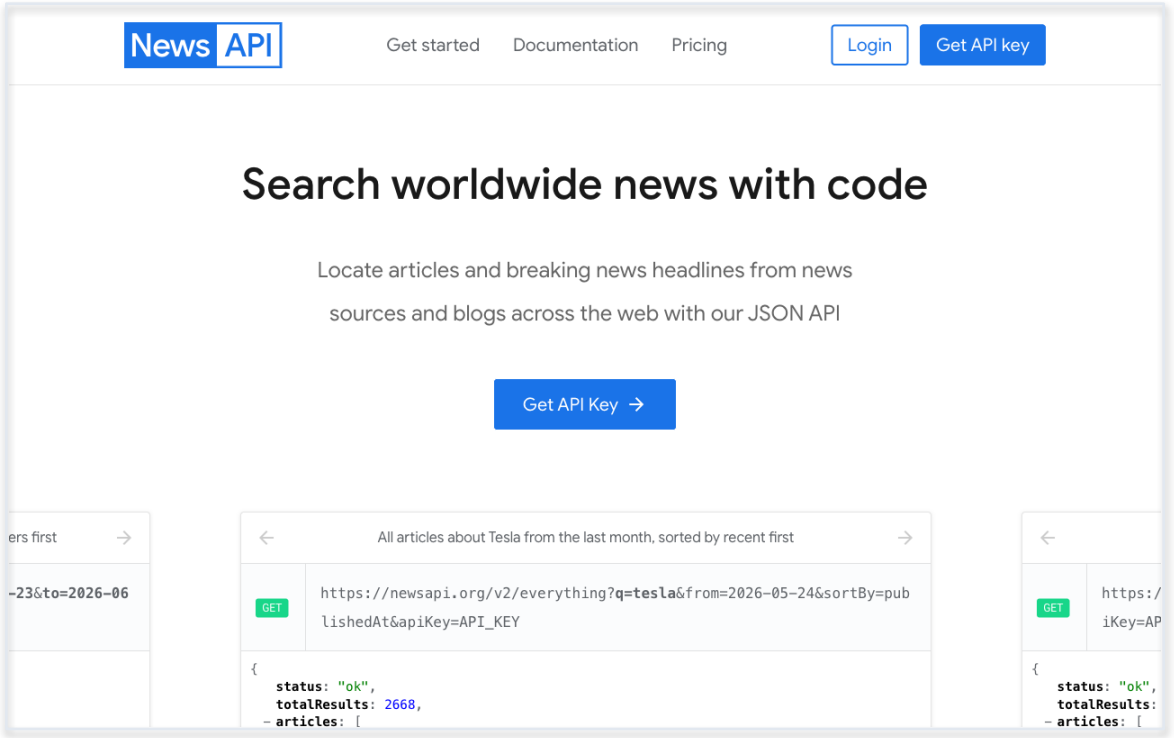
- Live stock / forex / crypto data over a simple REST API.
- Sign up (free Basic plan), then Account → API Keys.
- Paste the apikey into the three 'candles' HTTP nodes.
- Free tier: 8 calls/min — enough for the lab.



© 2026 Tertiary Infotech Academy Pte Ltd

NewsAPI — Headlines API

- Recent news articles for any search query.
- Register (free Developer plan) and copy your API key.
- Store it as a Query Auth credential (name = apiKey) on the news node.
- The agent reads sentiment from the headlines.



© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 4 · LAB WORKFLOW

Activity 6 — Finance APIs → Telegram (AI Day-Trading Agent)

Ask the bot about a stock: it pulls candles on three timeframes plus news, and the agent argues a Buy / Sell / Hold call.

Telegram Trigger
'Should I buy AAPL?'

▶

Code — Extract Ticker
AAPL

▶

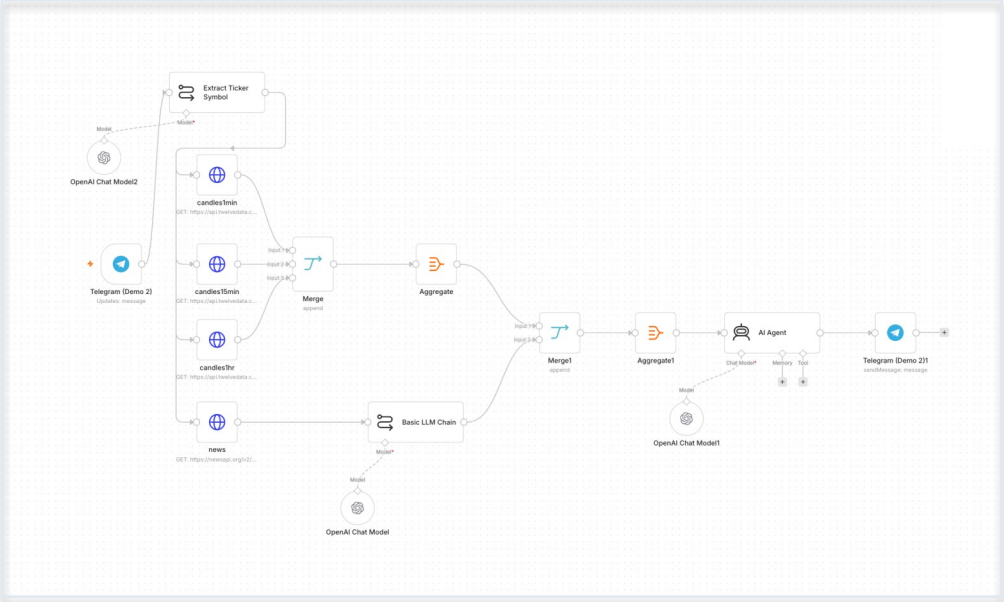
HTTP ×4
candles 1m / 15m / 1h + news

▶

AI Agent
technicals + sentiment → call

▶

Telegram — Send
recommendation + reasoning



ACT 6

Finance APIs → Telegram (AI Day-Trading Agent)

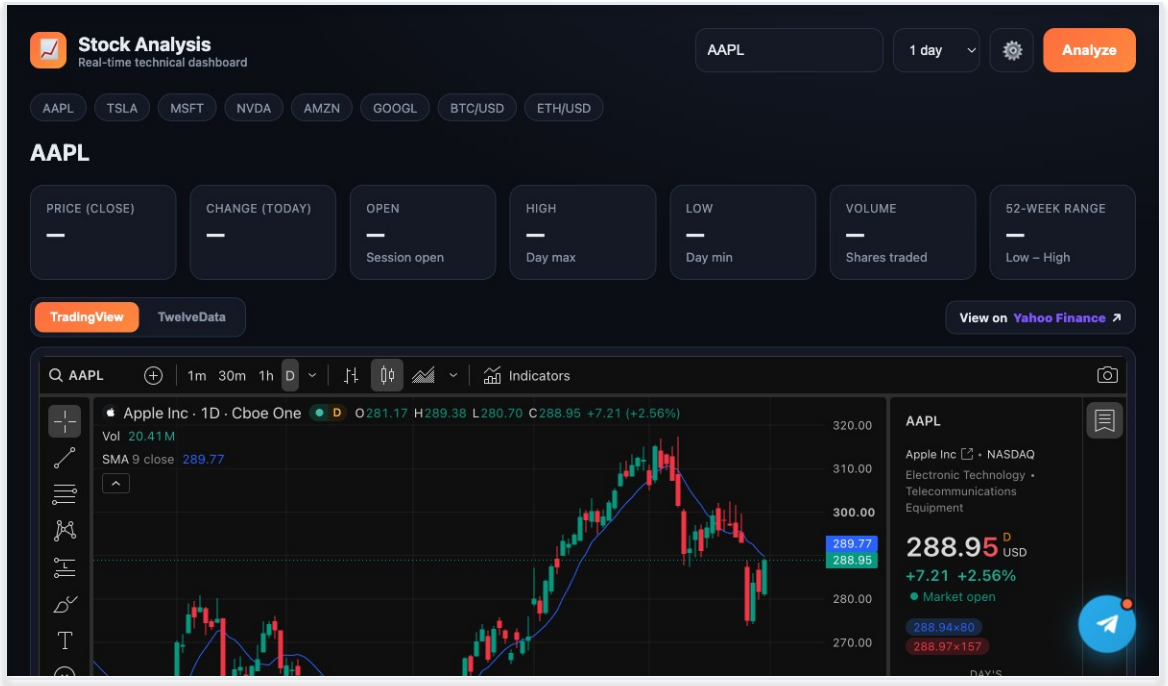
Twelve Data + NewsAPI keys · imported flow, re-select credentials · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

ACT 6 · THE WEBSITE

Activity 6 — The Stock Analysis Dashboard

- A real-time dashboard with a live TradingView candlestick chart.
- Quote stats (price, change, OHLC, volume, 52-week range) from Twelve Data.
- A floating Telegram chat widget opens the AI day-trading agent.
- Save your Twelve Data key + bot username in the Settings panel.



© 2026 Tertiary Infotech Academy Pte Ltd

WRAP-UP

End of Day 2 — Recap

- You exposed an AI agent to a public website via a webhook (Production URL + CORS).
- You pulled live market and news data through APIs into a Telegram agent.
- Tomorrow: RAG at production scale, human-in-the-loop, guardrails, and the capstone.

© 2026 Tertiary Infotech Academy Pte Ltd

Lunch Break

1 hour

© 2026 Tertiary Infotech Academy Pte Ltd

05

TOPIC 5

Retrieval-Augmented Generation

Why retrieval · embeddings and vector stores · chunking and top_k · production vector DBs

Applied in Activities 7a and 7b

© 2026 Tertiary Infotech Academy Pte Ltd

Why Retrieval, Rather Than a Bigger Prompt

You could paste all your documents into the instruction — until you can't.

Everything in the prompt

Beyond a point the model ignores the middle. Every question pays tokens for ALL documents. Add document #41 and quality quietly drops.

Retrieve, then generate

Find the two or three chunks that resemble the question and give the model ONLY those. Cost stays flat as the library grows.

Question
embed it

▶

Search the store
nearest chunks

▶

Paste into the prompt
only what matched

▶

Grounded answer
cites what it saw

RAG = retrieval + generation. The retrieval decides whether the generation can possibly be right.

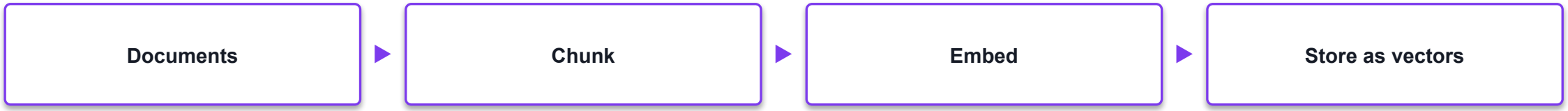
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 5 · RAG

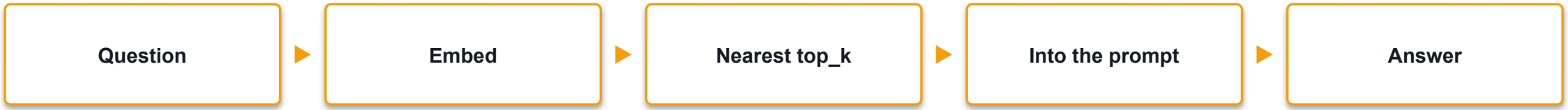
The RAG Pipeline — Ingest Once, Retrieve Every Question

Two phases: ingestion happens once per document; retrieval happens on every question.

INGESTION — once



RETRIEVAL — every question



Chunk

How a document is split — here one brochure = one chunk.

Embedding

Text as a list of numbers; similar meaning → nearby vectors.

Similarity

Nearness of two vectors — how 'related' two texts are.

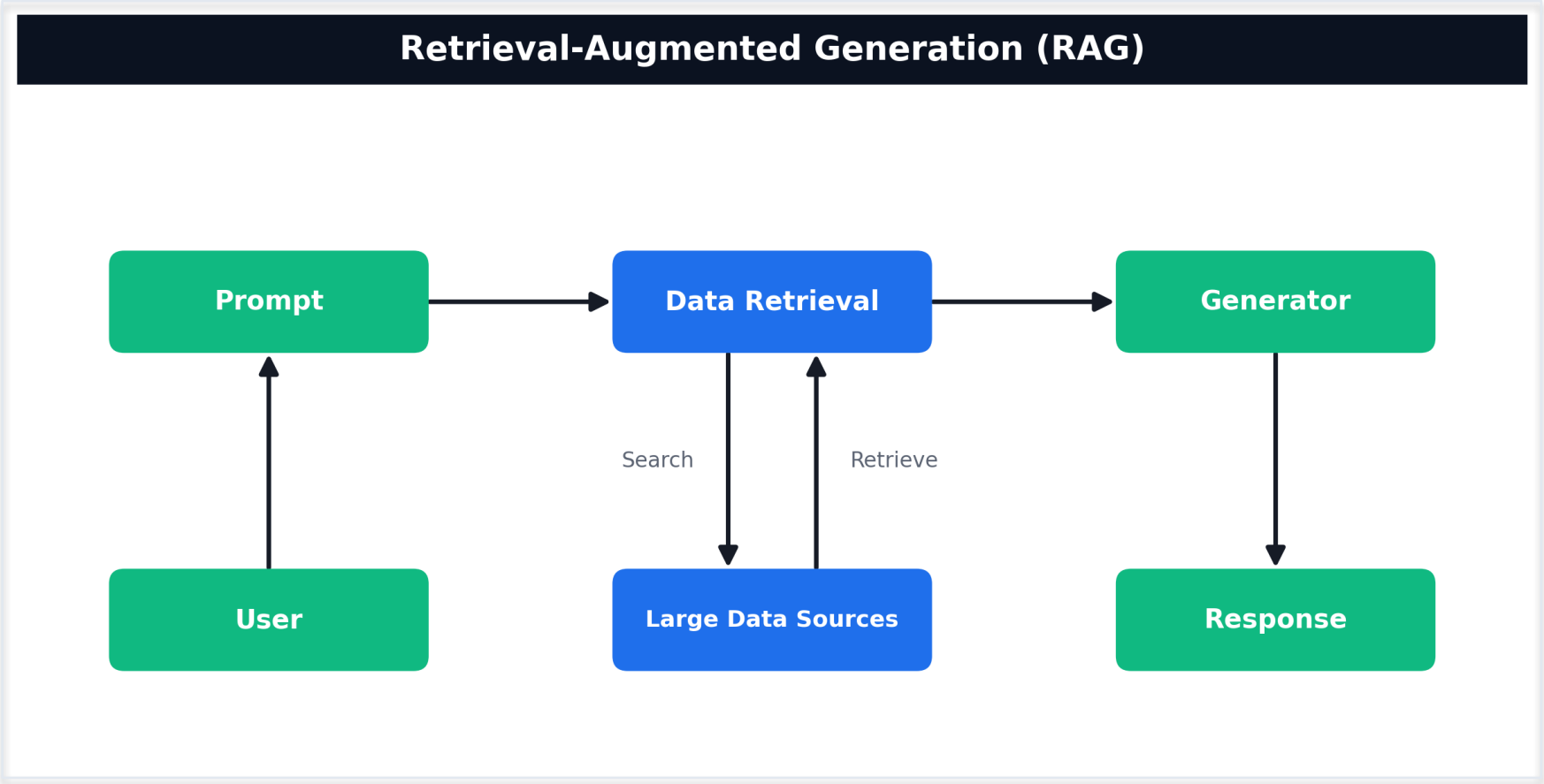
top_k

How many chunks come back per question — 3 here.

Four words — chunk, embedding, similarity, top_k — explain every RAG behaviour you will see.

© 2026 Tertiary Infotech Academy Pte Ltd

How RAG Works



User → Prompt → Data Retrieval (search/retrieve over your data sources) → Generator → Response

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 5 · RAG

Vector Databases — Where the Embeddings Live

In-memory for demos; a hosted store for anything that must survive a restart.

	Supabase (pgvector)	Pinecone	Qdrant
What it is	Postgres + extension	Managed SaaS index	Open-source vector DB
Setup	SQL table + function	Create an index — no schema	Docker or Qdrant Cloud
Best for	Teams already on Postgres	Zero-ops, scales fast	Full control, self-hosted
Course embedding	OpenAI 1536-dim	Gemini 3072-dim	OpenAI 1536-dim

THE DIMENSION RULE

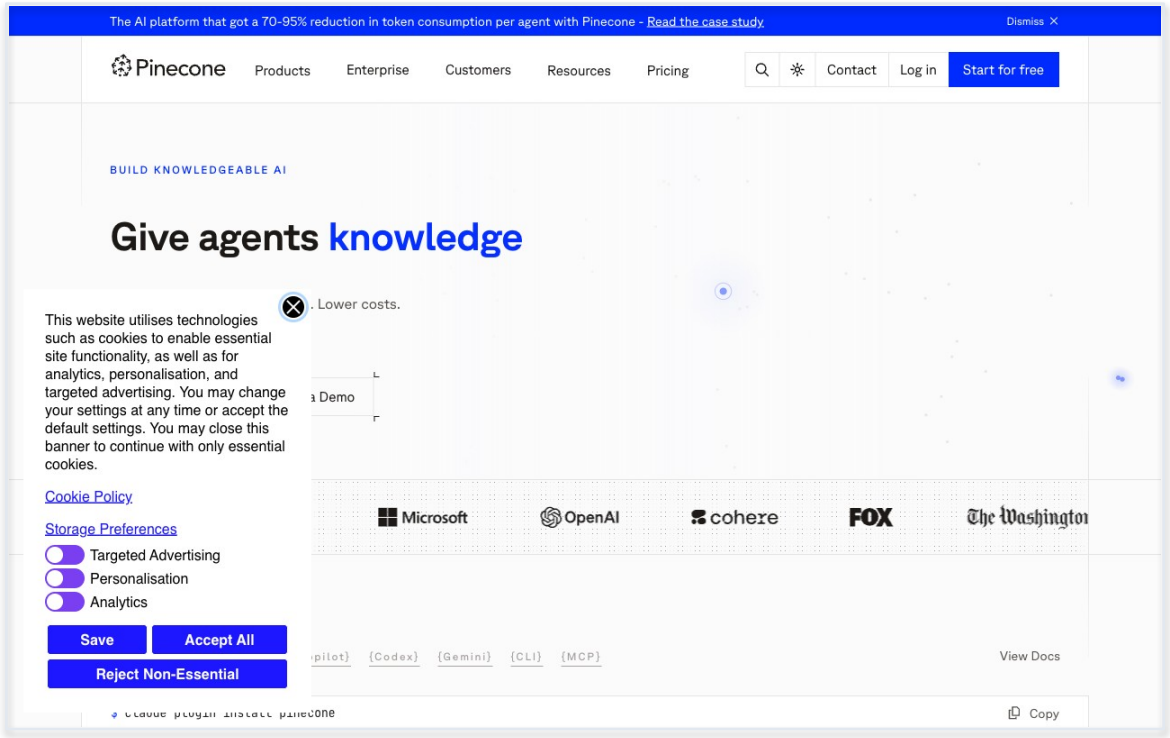
The store's dimension MUST equal the embedding model's output — OpenAI text-embedding-3-small = 1536, Gemini gemini-embedding-001 = 3072. Mismatched dimensions are the #1 cause of failed inserts.

Swap the store, keep the RAG flow — but ingestion and retrieval must use the SAME embedding model.

© 2026 Tertiary Infotech Academy Pte Ltd

Pinecone — Example Managed Store

- Pinecone is a managed (cloud) vector database for RAG.
- Free 'Starter' tier is enough for this lab.
- Create an index: Dimensions = 3072 (Gemini), Metric = cosine.
- Upload and retrieval must use the same namespace.



© 2026 Tertiary Infotech Academy Pte Ltd

Activity 7a — RAG Chatbot: Upload a PDF, Ask in Telegram

The full RAG loop, no code: a web page ingests your IT-Support FAQ; a Telegram bot answers only from that document.

Web Uploader
PDF text extracted in browser

▶

Upload Webhook
POST to n8n

▶

Gemini Embeddings
chunk + embed

▶

Vector Store
in-memory; replaced per upload

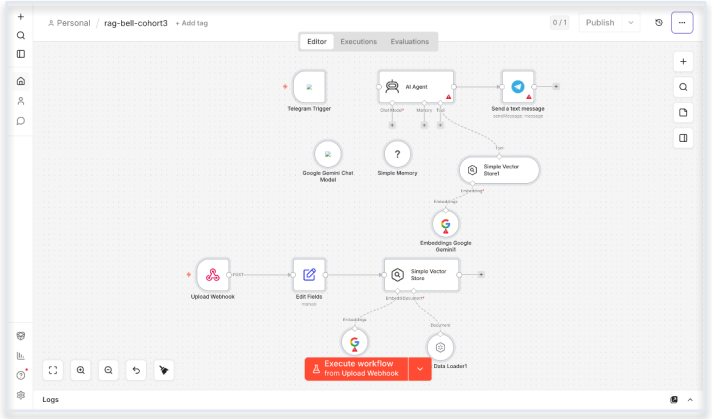
Telegram Trigger
'How do I reset my password?'

▶

AI Agent
knowledge_base retriever tool

▶

Telegram — Send
answers only from the document



ACT 7a

RAG Chatbot: Upload a PDF, Ask in Telegram

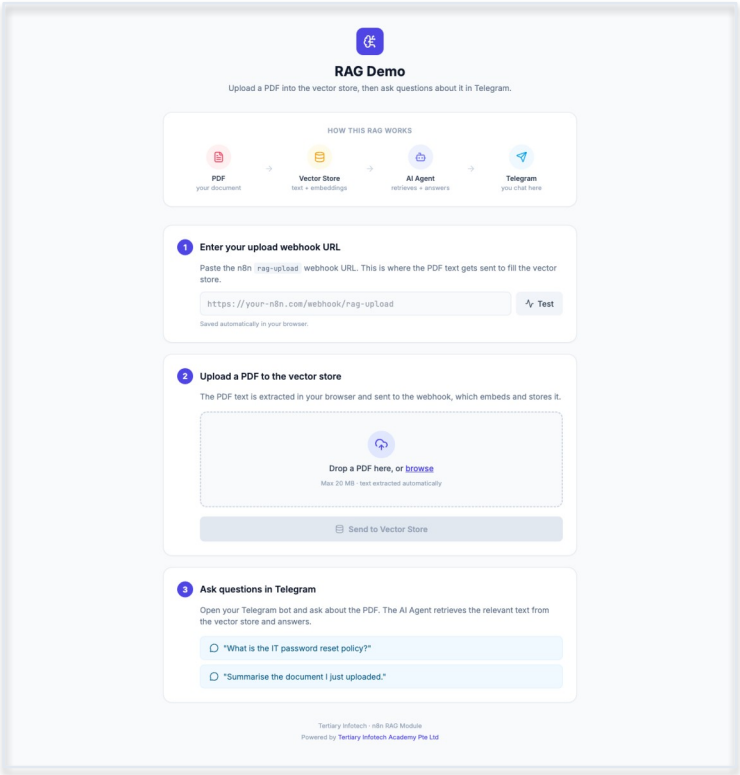
Gemini credential on 3 nodes · off-topic questions are refused · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

ACT 7a · INGESTION

Activity 7a — The Upload Page

- A single-page uploader extracts PDF text in the browser with PDF.js.
- Paste your n8n rag-upload webhook URL and click Test.
- Drop the IT-Support FAQ PDF and Send to Vector Store to ingest it.
- No build step — just open the HTML file.



© 2026 Tertiary Infotech Academy Pte Ltd

Activity 7b — Ingestion: 20 Brochures into a Vector DB

Cook & Bake Academy: load 20 course brochures from Google Drive into your chosen vector database — run once.

Manual Trigger
run once

▶

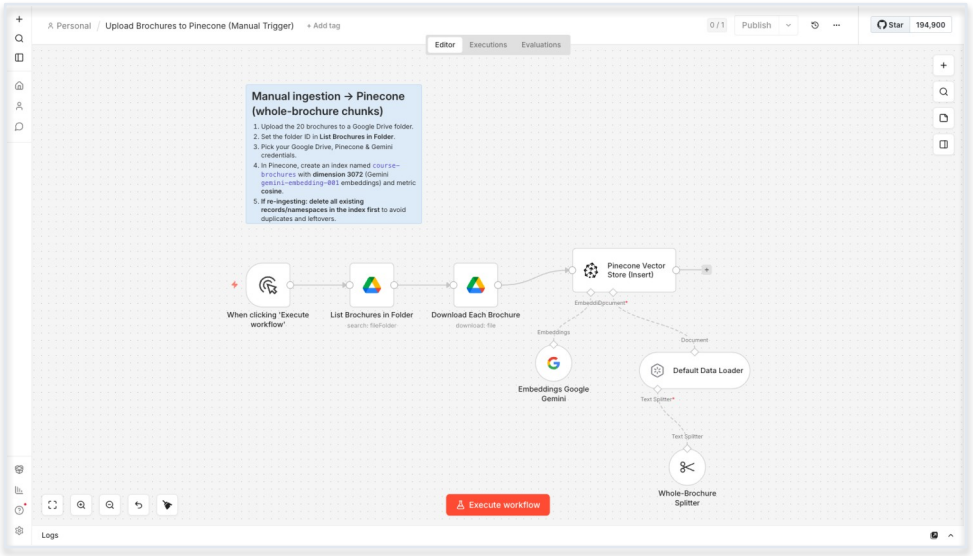
Google Drive
list + download 20 brochures

▶

Split + Embed
1 brochure = 1 chunk · Gemini 3072-d

▶

Vector Store — Insert
index: course-brochures



Shown: Pinecone ingestion. Supabase & Qdrant flows are identical except the store node + OpenAI 1536-d embeddings.

ACT 7b

Ingestion: 20 Brochures into a Vector DB

Choose Supabase / Pinecone / Qdrant · 20 vectors after the run · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 5 · LAB WORKFLOW

Activity 7b — CX Agent: Grounded Support Chatbot

The answering half: the website chat widget POSTs each question; the agent retrieves matching brochures and answers grounded.

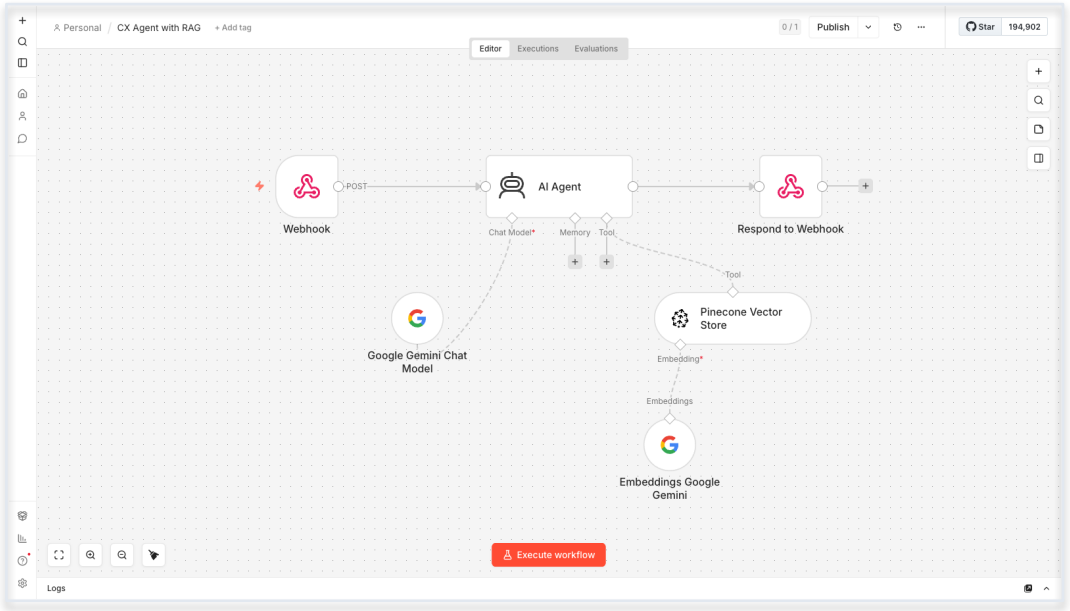
Webhook — POST
website chat widget

▶

AI Agent
Pinecone retriever tool + Gemini

▶

Respond to Webhook
grounded answer to the widget



ACT 7b

CX Agent: Grounded Support Chatbot

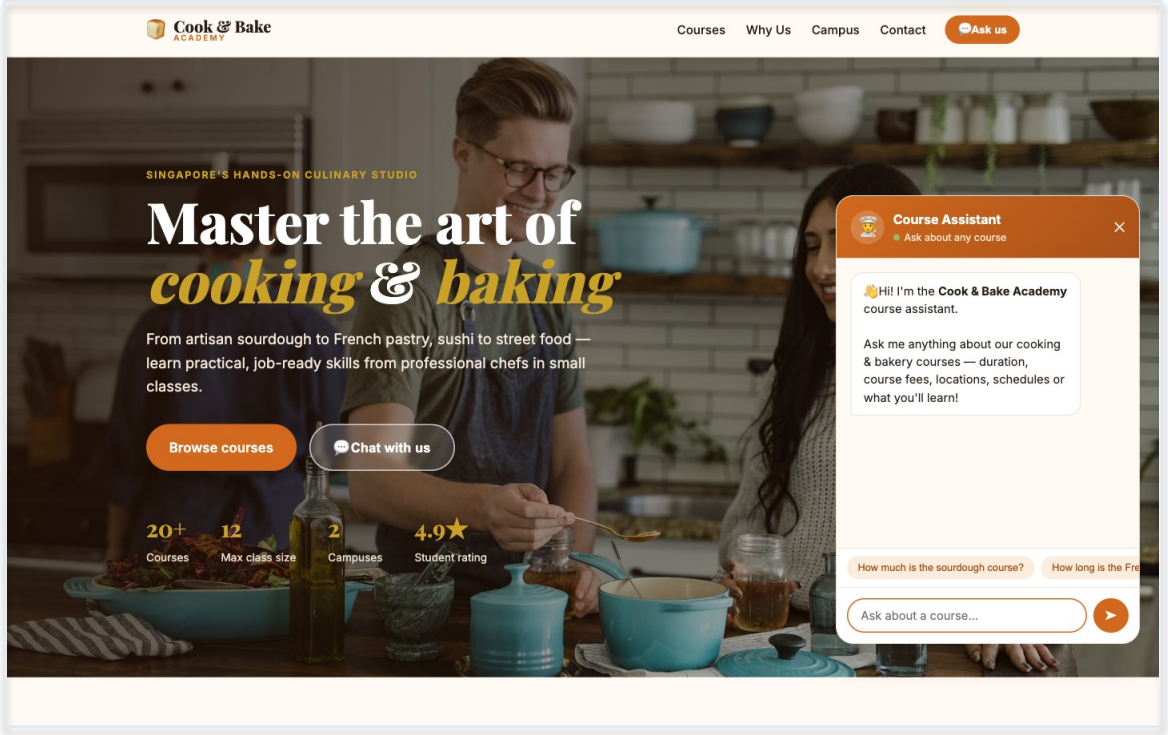
Same index, same embeddings, same namespace as ingestion · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

ACT 7b · THE WEBSITE

Cook & Bake Academy — Support Chatbot

- The Cook & Bake Academy site with the course assistant open.
- Visitors ask about any course — duration, fee, location, schedule.
- Answers are grounded in the 20 ingested brochures.
- Set WEBHOOK_URL in script.js to your n8n CX Agent Production URL.

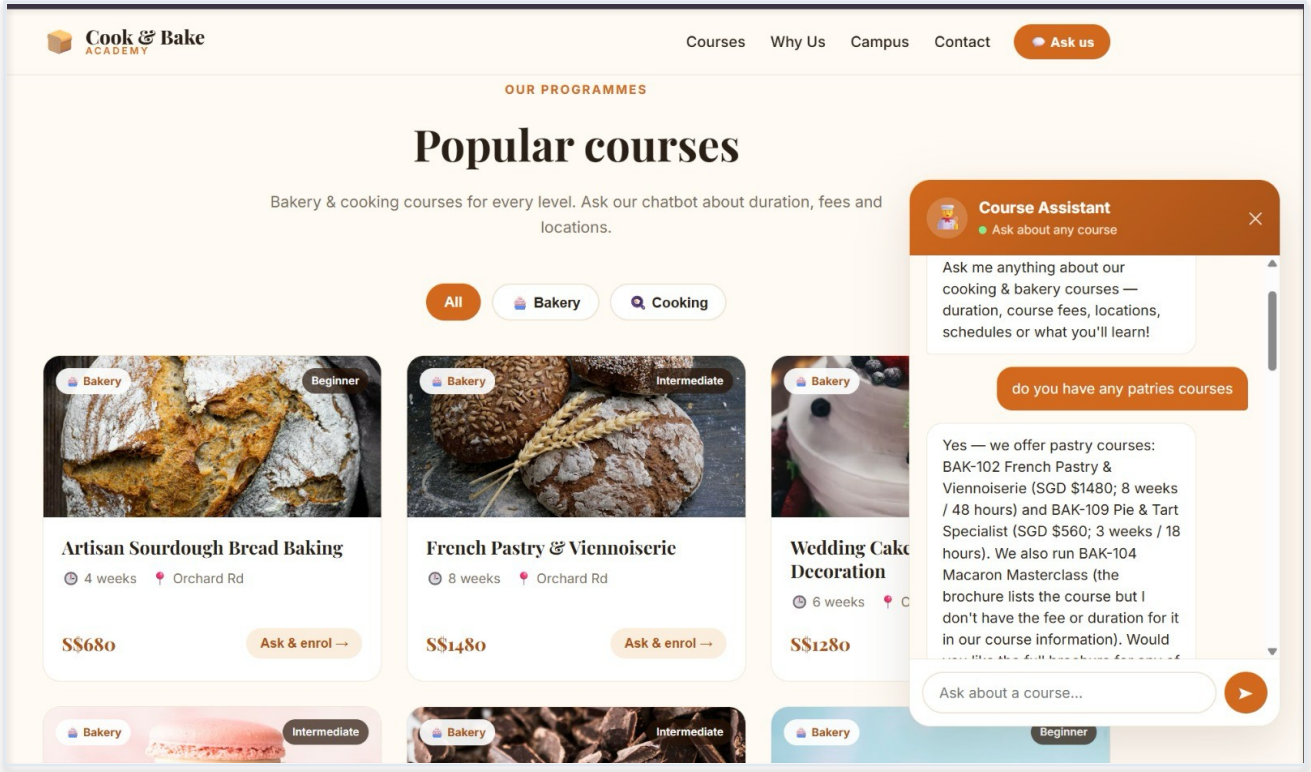


© 2026 Tertiary Infotech Academy Pte Ltd

ACT 7b · THE WEBSITE

Activity 7b — The Course Assistant in Action

A real conversation on the Cook & Bake Academy site. Every fee in the reply came from a retrieved brochure.



1 A typo still finds it

The visitor typed 'patries'. Embeddings match meaning, not spelling, so the pastry brochures BAK-102 and BAK-109 still come back.

2 Facts quoted, not invented

Fees and durations are quoted exactly: BAK-102 is SGD $1480 for 8 weeks / 48 hours — the same S$1480 shown on the course card behind the chat.

3 It admits a gap

It did not give BAK-104's fee and said so instead of guessing. The brochure does list SGD $420, so check what the tool returned: that is a retrieval gap.

'I don't have that' is grounding working — fix gaps in retrieval, never by loosening the prompt.

© 2026 Tertiary Infotech Academy Pte Ltd

Tea Break

15 minutes

© 2026 Tertiary Infotech Academy Pte Ltd

06

TOPIC 6

Security, Human-in-the-Loop & Guardrails

Where the human belongs · approval gates · input and output guardrails

Applied in Activities 8a, 8b and 8c

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 6 · SECURITY & GOVERNANCE

Human In, On, and Out of the Loop

Three arrangements people use interchangeably — and they are not the same thing.

Pattern	Who decides	Who acts	Can AI proceed alone?
Human IN the loop	AI proposes, human decides	Human approves, system acts	NO — it blocks
Human ON the loop	AI decides and acts	Human monitors, can intervene	Yes — reviewed after
Human OUT of the loop	AI decides and acts	AI acts	Yes — nobody checks

WHERE THIS COURSE PUTS THE HUMAN

Activity 8a — the manager approves leave before any email goes to the employee (IN the loop). The dashboard (8b) and chatbot (8c) run alone, but the chatbot is fenced by guardrails on both sides.

Ask who pays for the mistake — that answer sizes the agent's authority and places the human.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 6 · SECURITY & GOVERNANCE

How an Approval Suspends a Running Flow

n8n's Send and Wait for Response physically pauses the workflow until a person clicks.

Request submitted
employee applies for leave

▶

Send & Wait
manager gets Approve / Decline email

▶

The run SUSPENDS
minutes or days — it waits

▶

A person responds
one click in the email

▶

IF reads the outcome
approved → notify · declined → notify

Outcome = Approve

Send the approval email, update the record, continue the business process.

Outcome = Decline

Send the rejection with the manager's comments — never silence.

Ack first

The portal gets an instant 'received' response BEFORE the wait — the browser never hangs on a manager.

A pause that is still paused tomorrow is a real gate — that is the deliverable of Activity 8a.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 6 · SECURITY & GOVERNANCE

Guardrails — Two Gates Around the Agent

An LLM checks every message in, and every reply out — the agent only sees and says what passes.



1 Input (pre) guardrail blocks

Prompt injection ('ignore previous instructions'), jailbreaks, requests for another person's salary or private data.

2 Output (post) guardrail blocks

Salary figures, NRIC, credentials, system-instruction leaks — even when the agent was tricked into producing them.

Instructions say what the agent SHOULD do; guardrails enforce what it MUST NOT — you need both.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 6 · LAB WORKFLOW

Activity 8a — Leave Application & Manager Approval (HITL)

The employee applies on the portal; the flow acks, then suspends until the manager clicks Approve or Decline in email.

Webhook — POST
hr-leave-apply

▶

Respond — Ack
portal gets instant OK

▶

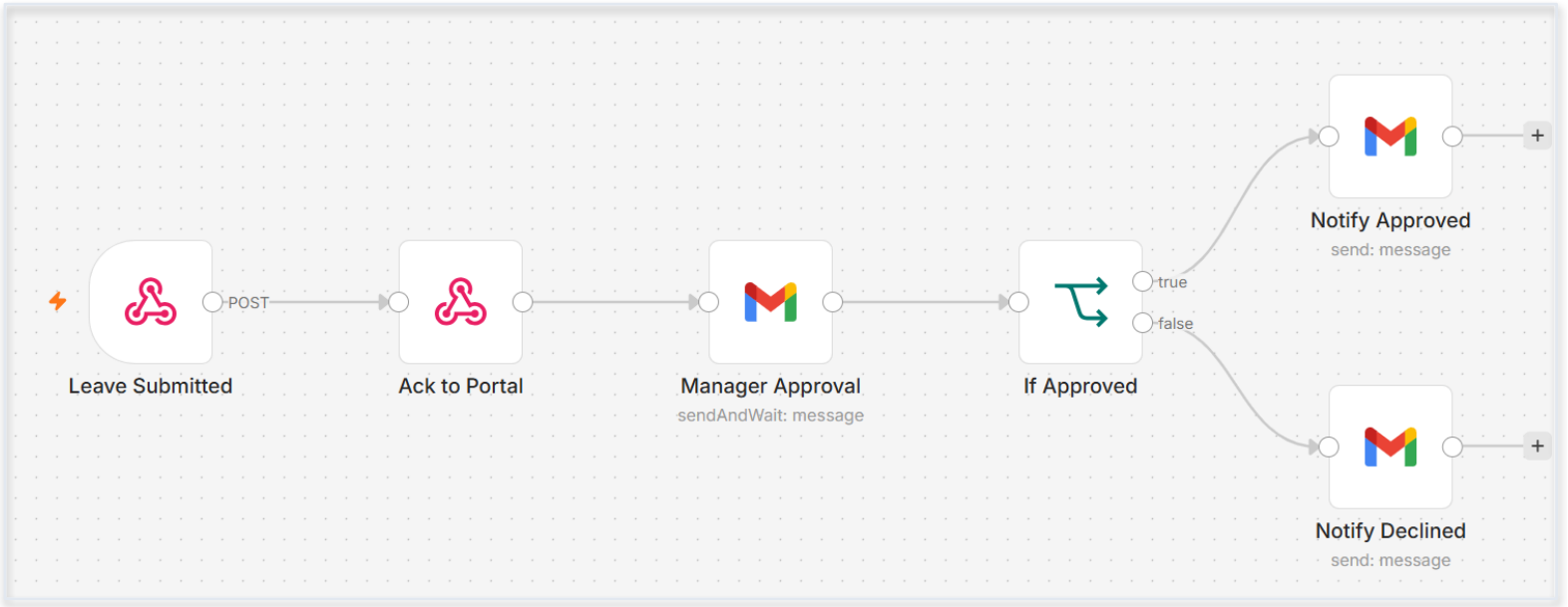
Gmail — Send & Wait
manager: Approve / Decline

▶

IF
approved = true ?

▶

Gmail
notify the employee



ACT 8a **Leave Application & Manager Approval (HITL)**

Human-in-the-loop gate · run suspends until the click · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 6 · LAB WORKFLOW

Activity 8b — HR Dashboard Data (GET Webhook)

A GET webhook powers the portal's Dashboard tab — leave balance and history returned as JSON to the browser.

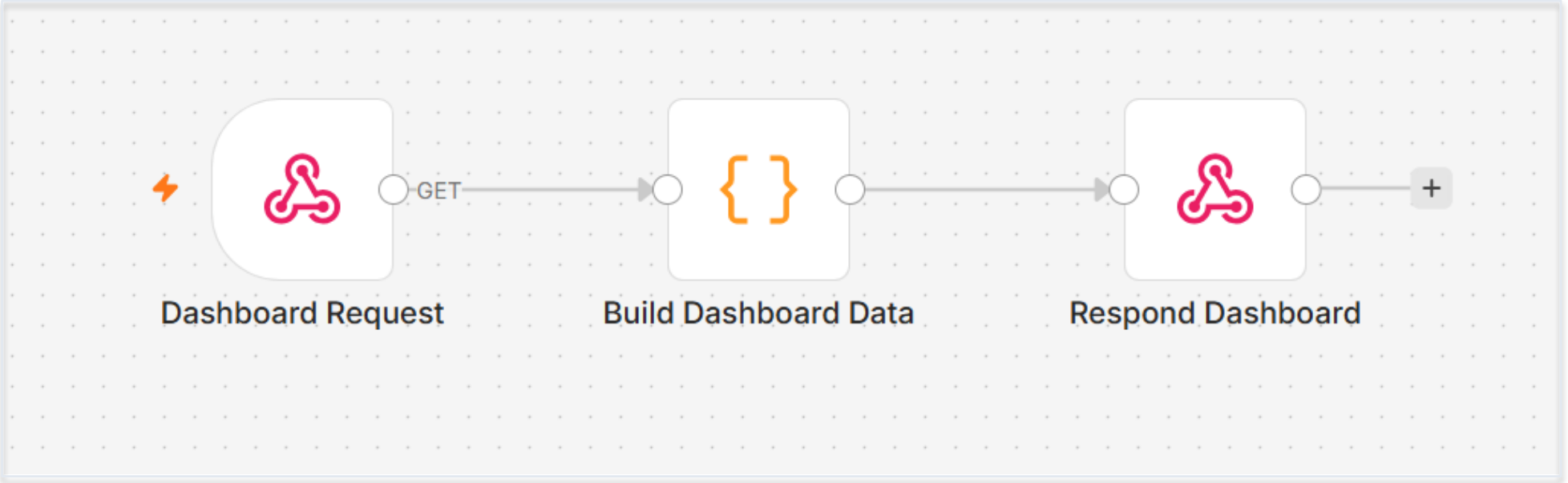
Webhook — GET
hr-dashboard?email=…

▶

Code
build the leave-data JSON

▶

Respond to Webhook
JSON back to the portal



ACT 8b

HR Dashboard Data (GET Webhook)

GET query read from $json.query.email · CORS = * · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 6 · LAB WORKFLOW

Activity 8c — AI Chatbot with Input & Output Guardrails

HR Buddy answers general HR questions — every message passes the input gate, every reply passes the output gate.

Webhook — POST
hr-chat

▶

Input Guardrail
LLM: ALLOW / BLOCK

▶

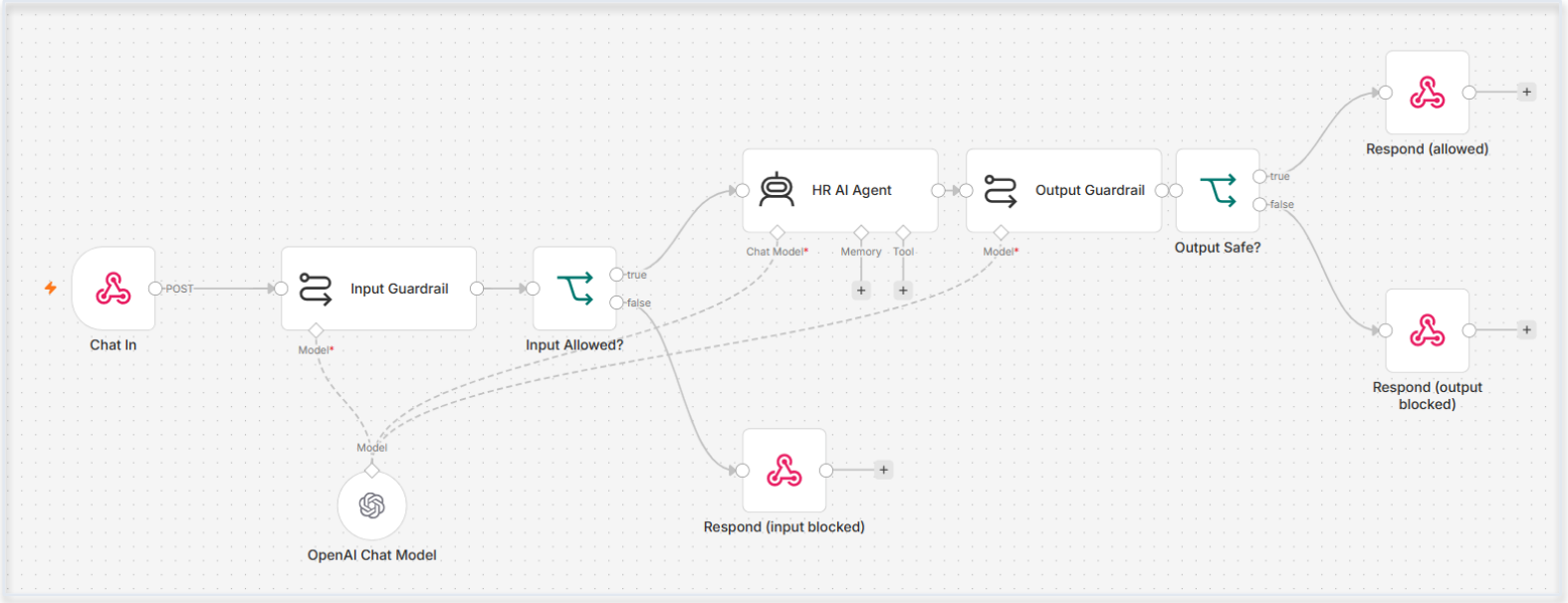
HR AI Agent
leave · payroll · claims

▶

Output Guardrail
LLM: SAFE / LEAK

▶

Respond
reply or blocked message



ACT 8c

AI Chatbot with Input & Output Guardrails

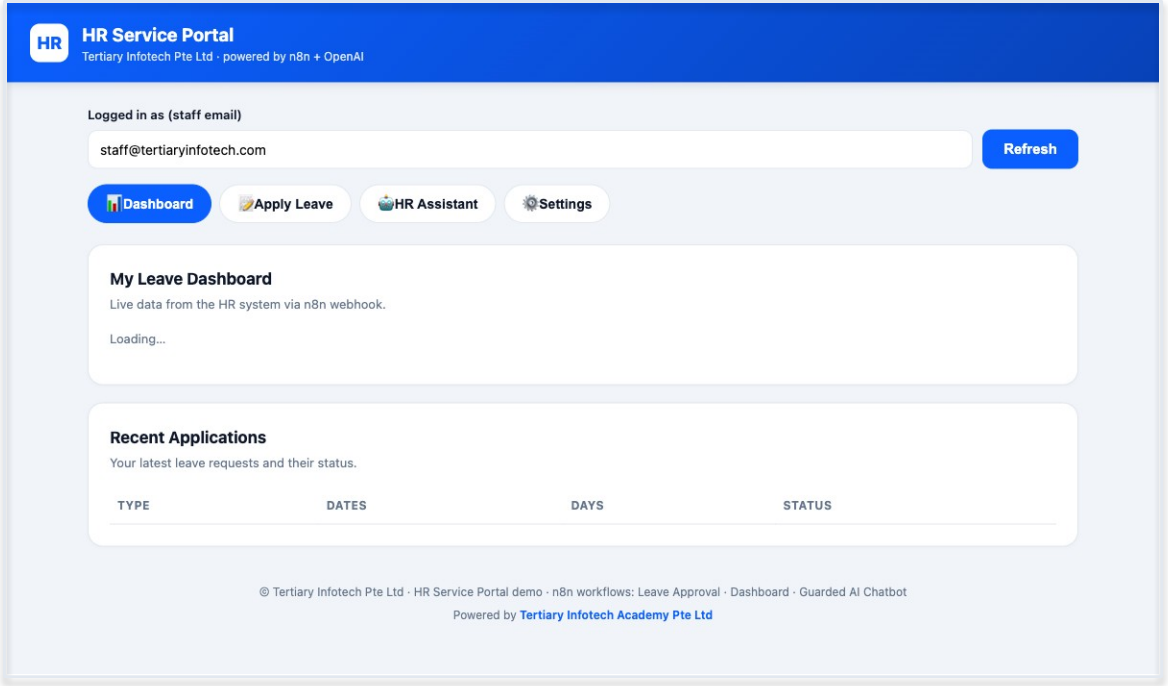
Test it by trying to break it — the injection probe must be blocked · steps in the Learner Guide

© 2026 Tertiary Infotech Academy Pte Ltd

ACT 8 · THE WEBSITE

Activity 8 — The HR Service Portal

- One portal with four tabs: Dashboard, Apply Leave, HR Assistant, Settings.
- Dashboard pulls live leave balance & history from the GET webhook (8b).
- Apply Leave triggers the human-in-the-loop approval chain (8a).
- HR Assistant is the guarded AI chatbot — input & output guardrails (8c).



© 2026 Tertiary Infotech Academy Pte Ltd

Lunch Break

1 hour

© 2026 Tertiary Infotech Academy Pte Ltd

07

TOPIC 7

Mini Capstone Project

Describe it · n8n Assistant builds it · review · present

Everything from Activities 1–8 — built with AI-assisted workflow creation

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 7 · MINI CAPSTONE

Your Capstone — Let n8n Assistant Build It

In small groups, pick one end-to-end business process, describe it to n8n Assistant, then review, test and own the workflow it builds.

Pick a process
one painful, repeatable job

▶

Write the prompt
trigger → steps → output

▶

n8n Assistant builds
plan, nodes, wiring

▶

You review & connect
credentials, Gateway credits

▶

Test, add a control
guardrail or approval

▶

Present
demo + 3–5 min talk

1 A trigger

Form, webhook, Telegram or schedule — who or what starts your process?

2 An AI agent

With at least one tool or a RAG source — the judgement in the middle.

3 Data or storage

An API you call, or storage you write — Sheets, a vector DB, a Data Table.

4 A control

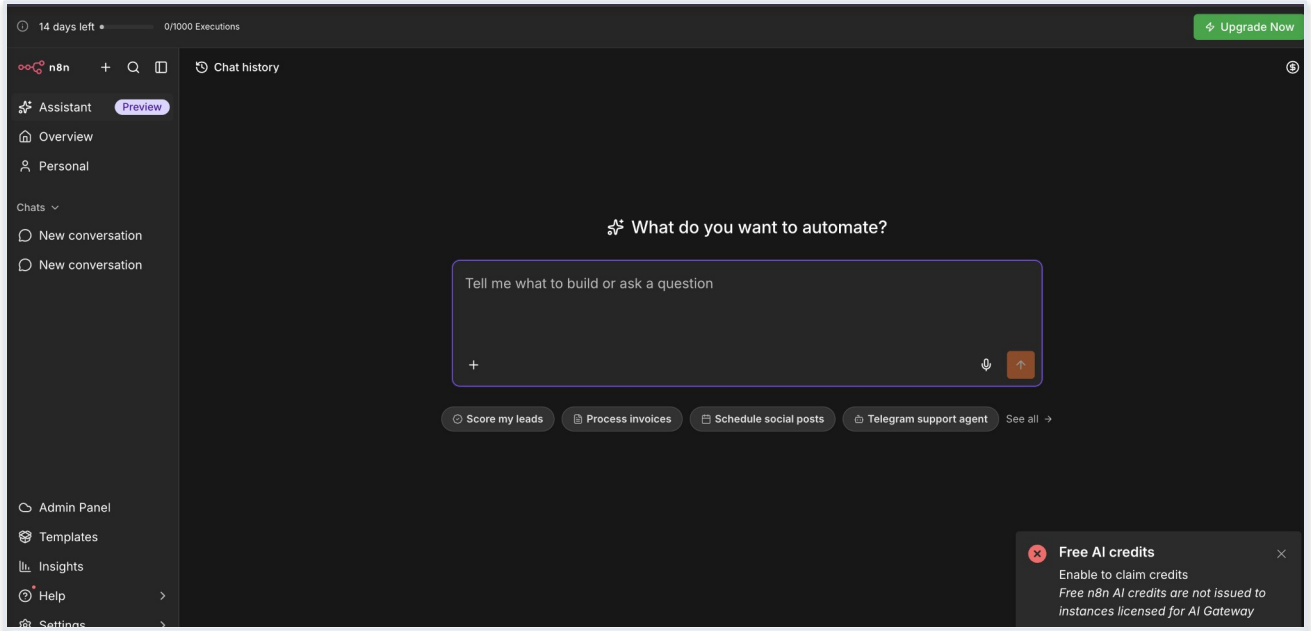
A guardrail or a human-in-the-loop step — decide where the human belongs and defend it.

Deliverables: the prompt(s) you used, a working workflow (.json), a short demo and a 3–5 minute presentation.

© 2026 Tertiary Infotech Academy Pte Ltd

Step 1: Describe Your Process to n8n Assistant

Open Assistant in the left sidebar (or Build with AI on an empty canvas) and answer 'What do you want to automate?'



1 Say what starts it

Name the trigger and its inputs — 'a form where I enter a topic, audience and length'.

2 List the steps in order

Number them. One step per line becomes one node or one AI step on the canvas.

3 Say where the human decides

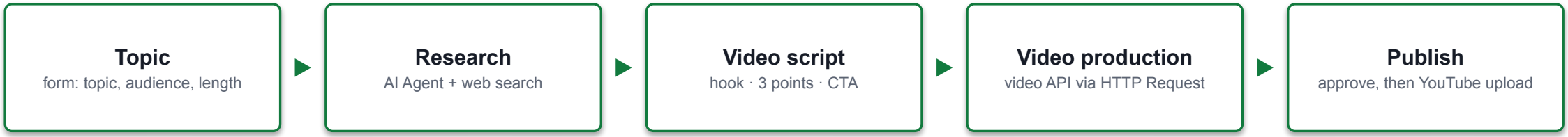
Ask for an approval gate before anything is published, sent or paid.

A numbered prompt gives a numbered workflow — the clearer the steps, the closer the first draft.

© 2026 Tertiary Infotech Academy Pte Ltd

Example: An End-to-End Video Creation Workflow

One prompt turns a topic into a published video — research, script, production and publishing, with a human approval gate.



THE PROMPT

Create an end-to-end workflow that turns a topic into a published YouTube video:
1. Start from a form where I enter a video topic, target audience and length (60–90 seconds).
2. Research the topic on the web and summarise 5 key facts with their sources.
3. Write a video script from the research: a hook, 3 key points and a call to action, under 200 words.
4. Send the script to a video-generation API with an HTTP Request node, and wait until the video is ready.
5. Email me the script and the video link, and wait for my approval.
6. If approved, upload the video to YouTube with a title, description and tags;
   if rejected, log my feedback in a Google Sheet.

Use Gateway credits for the AI models. Ask me before publishing.

THEN YOU

- Review the plan before it builds
- Add your video API key as a credential (e.g. HeyGen, Runway or Google Veo)
- Connect Gmail, Google Sheets and YouTube
- Test with a real topic — approve the draft video

The AI writes and produces; the workflow runs it — and nothing is published until a person approves it.

© 2026 Tertiary Infotech Academy Pte Ltd

More Example Prompts for Your Capstone

Start from one of these, or write your own. Each names the trigger, numbers the steps and puts a human at the risky step.

Weekly newsletter

1. Every Monday at 9am, search the web for this week's news on <your industry>.
2. Pick the 5 most relevant stories and write a short newsletter with a summary and link for each.
3. Email me the draft for approval; if approved, send it to the subscribers in my Google Sheet.

Invoice processing

1. When an email with a PDF invoice arrives in Gmail, extract the supplier, invoice number, date and amount.
2. Add a row to my Invoices Google Sheet.
3. If the amount is over $5,000, email the finance manager to approve before marking it 'ready to pay'.

Customer-support triage

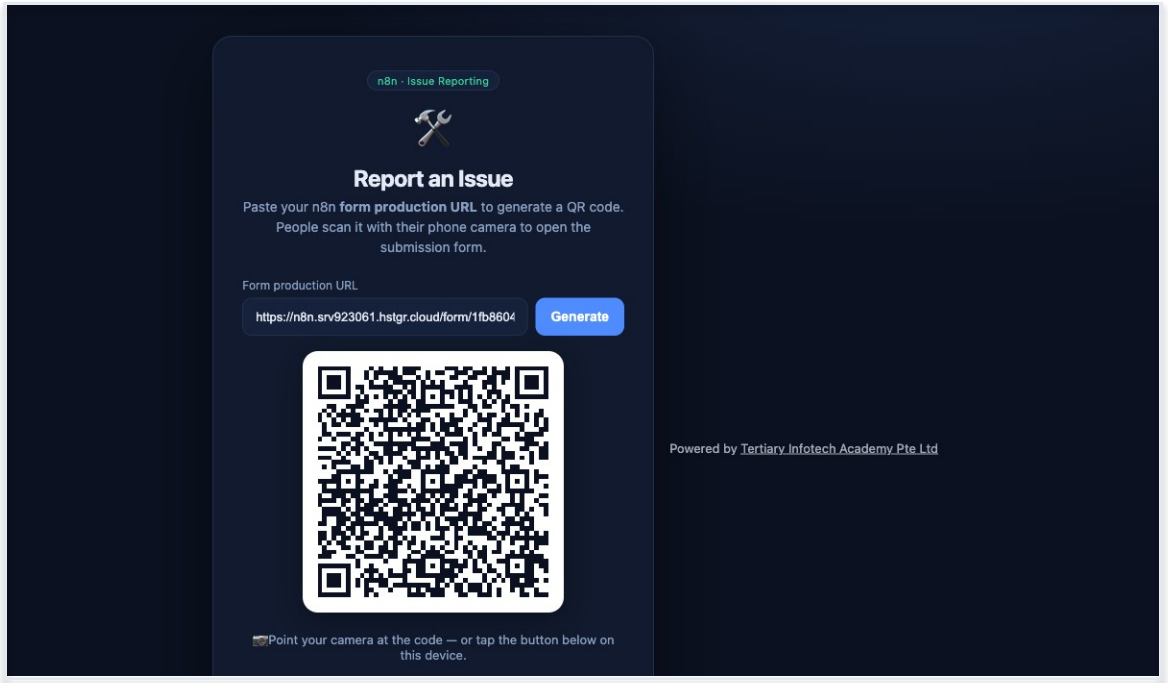
1. From a website chat webhook, classify each message as billing, technical or sales.
2. Answer FAQs from my policy documents (RAG); never invent a price.
3. Hand anything else to a human by posting it to Telegram, and log every conversation.

Then refine in the same chat: 'add an approval step', 'use Gateway credits', 'log every run to a Google Sheet'.

© 2026 Tertiary Infotech Academy Pte Ltd

Capstone Example — Issue Reporting

- Paste your n8n Form production URL to generate a QR code.
- People scan the QR with a phone to open the submission form.
- Submissions (with photos) flow into Postgres and a public gallery.
- A complete worked capstone you can adapt for your own project.



© 2026 Tertiary Infotech Academy Pte Ltd

WRAP-UP

Presentation & Assessment

- Present the problem, your design, and what you'd improve.
- Written Assessment (SAQ) — 1 hour · Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials.
- Remember to take the Assessment digital attendance (TRAQOM).

© 2026 Tertiary Infotech Academy Pte Ltd

FINAL ASSESSMENT

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

Thank You!

Keep your local n8n running and keep building your own agents.

Download all flows: github.com/tertiarycourses/TGS-2023035977-Agentic-AI-Automation-with-n8n

Powered by Tertiary Infotech Academy Pte Ltd · www.tertiarycourses.com.sg

© 2026 Tertiary Infotech Academy Pte Ltd